

Semiconductor Physics

Learning Objectives

- Understand the properties of semiconductors.
- Commonly used semiconductors in electronics.
- Recognise the effect of temperature on semiconductors.
- Understand the concept of hole current in semiconductors.
- Develop a strong understanding of p -type semiconductor and n -type semiconductor.
- Role of majority and minority charge carriers in n -type and p -type semiconductors.
- Develop an understanding of pn junction.
- Develop a strong understanding of forward bias and reverse bias operating mode of pn junction (i.e. semiconductor diode).
- Volt-Ampere (V - I) characteristics of pn junction.
- Understand the reverse breakdown voltage and knee voltage of pn junction.

Introduction

Certain substances like germanium, silicon, carbon *etc.* are neither good conductors like copper nor insulators like glass. In other words, the resistivity of these materials lies inbetween conductors and insulators. Such substances are classified as *semiconductors*. Semiconductors have some useful properties and are being extensively used in electronic circuits. For instance, *transistor*—a semiconductor device is fast replacing bulky vacuum tubes in almost all applications. Transistors are only one of the family of semiconductor devices ; many other semiconductor devices are becoming increasingly popular. In this chapter, we shall focus our attention on the different aspects of semiconductors.

3.1 Semiconductor

It is not easy to define a semiconductor if we want to take into account all its physical characteristics. However, generally, a semiconductor is defined on the basis of electrical conductivity as under :

A **semiconductor** is a substance which has resistivity (10^{-4} to $0.5 \Omega\text{m}$) inbetween conductors and insulators e.g. germanium, silicon, selenium, carbon etc.

The reader may wonder, when a semiconductor is neither a good conductor nor an insulator, then why not to classify it as a *resistance material* ? The answer shall be readily available if we study the following table :

S.No.	Substance	Nature	Resistivity
1	Copper	good conductor	$1.7 \times 10^{-8} \Omega\text{ m}$
2	Germanium	semiconductor	$0.6 \Omega\text{ m}$
3	Glass	insulator	$9 \times 10^{11} \Omega\text{ m}$
4	Nichrome	resistance material	$10^{-4} \Omega\text{ m}$

Comparing the resistivities of above materials, it is apparent that the resistivity of germanium (semiconductor) is quite high as compared to copper (conductor) but it is quite low when compared with glass (insulator). This shows that resistivity of a semiconductor lies inbetween conductors and insulators. However, it will be wrong to consider the semiconductor as a resistance material. For example, nichrome, which is one of the highest resistance material, has resistivity much lower than germanium. This shows that electrically germanium cannot be regarded as a conductor or insulator or a resistance material. This gave such substances like germanium the name of semiconductors.

It is interesting to note that it is not the resistivity alone that decides whether a substance is semiconductor or not. For example, it is just possible to prepare an alloy whose resistivity falls within the range of semiconductors but the alloy cannot be regarded as a semiconductor. In fact, semiconductors have a number of peculiar properties which distinguish them from conductors, insulators and resistance materials.

Properties of Semiconductors

- (i) A semiconductor is formed by covalent bonds.
- (ii) A semiconductor has crystalline structure.
- (iii) The resistivity of a semiconductor is less than that of an insulator but more than that of a conductor.
- (iv) Semiconductors have *negative temperature co-efficient of resistance* i.e. the resistance of a semiconductor decreases with the increase in temperature and *vice-versa*. For example, germanium is actually an insulator at low temperatures but it becomes a good conductor at high temperatures.
- (v) When a suitable metallic impurity (e.g. arsenic, gallium etc.) is added to a semiconductor, its current conducting properties change appreciably. This property is most important and is discussed later in detail.

3.2 Bonds in Semiconductors

The atoms of every element are held together by the bonding action of valence electrons. This bonding is due to the fact that it is the tendency of each atom to complete its last orbit by acquiring 8 electrons in it. However, in most of the substances, the last orbit is incomplete i.e. the last orbit does not have 8 electrons. This makes the atom active to enter into bargain with other atoms to acquire 8 electrons in the last orbit. To do so, the atom may lose, gain or share valence electrons with other atoms. In semiconductors, bonds are formed by sharing

of valence electrons. Such bonds are called *co-valent bonds*. In the formation of a co-valent bond, each atom contributes equal number of valence electrons and the contributed electrons are shared by the atoms engaged in the formation of the bond.

Fig. 3.1 shows the co-valent bonds among germanium atoms. A germanium atom has *4 valence electrons. It is the tendency of each germanium atom to have 8 electrons in the last orbit. To do so, each germanium atom positions itself between four other germanium atoms as shown in Fig. 3.1 (i). Each neighbouring atom shares one valence electron with the central atom. In this business of sharing, the central atom completes its last orbit by having 8 electrons revolving around the nucleus. In this way, the central atom sets up co-valent bonds. Fig. 3.1 (ii) shows the bonding diagram.

The following points may be noted regarding the co-valent bonds :

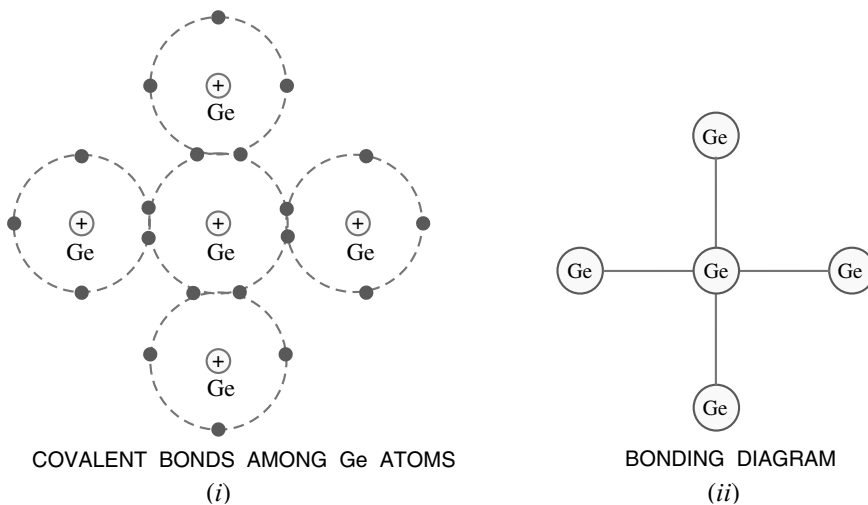


Fig. 3.1

(i) Co-valent bonds are formed by sharing of valence electrons.

(ii) In the formation of co-valent bond, each valence electron of an atom forms direct bond with the valence electron of an adjacent atom. In other words, valence electrons are associated with particular atoms. For this reason, valence electrons in a semiconductor are not free.

3.3 Crystals

A substance in which the atoms or molecules are arranged in an orderly pattern is known as a *crystal*. All semi-conductors have crystalline structure. For example, referring to Fig. 3.1, it is clear that each atom is surrounded by neighbouring atoms in a repetitive manner. Therefore, a piece of germanium is generally called germanium crystal.

3.4 Commonly Used Semiconductors

There are many semiconductors available, but very few of them have a practical application in electronics. The two most frequently used materials are *germanium* (Ge) and *silicon* (Si). It is because the energy required to break their co-valent bonds (*i.e.* energy required to release

* A germanium atom has 32 electrons. First orbit has 2 electrons, second 8 electrons, third 18 electrons and the fourth orbit has 4 electrons.

Therefore, the band gap is 2.1 eV. That is, the minimum energy E required to push an electron from the valence band into the conduction band is 2.1 eV. Hence the minimum energy required to create a hole-electron pair is 2.1 eV.

Example 3.7. In a pure semiconductor, the number of conduction electrons is 6×10^{19} per cubic metre. How many holes are there in a sample of size $1 \text{ cm} \times 1 \text{ cm} \times 1 \text{ mm}$?

Solution. In a pure semiconductor, hole density is equal to the electron density. Therefore, in the given semiconductor, hole density $= 6 \times 10^{19}/\text{m}^3$. Now the volume of the given sample $= 10^{-2} \times 10^{-2} \times 10^{-3} \text{ m}^3 = 10^{-7} \text{ m}^3$. Therefore, number of holes in the given sample

$$= 6 \times 10^{19} \times 10^{-7} = 6 \times 10^{12}$$

Example 3.8. Determine the number density of donor atoms which have to be added to an intrinsic germanium semiconductor to produce an n -type semiconductor of conductivity $5\Omega^{-1} \text{ cm}^{-1}$, given that mobility of conduction electrons in n -type Ge is $3900 \text{ cm}^2/\text{Vs}$. Neglect the contribution of holes to conductivity.

Solution. The conductivity of a semiconductor is given by ;

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

Neglecting the contribution of holes to conductivity, we have,

$$\sigma = e n_e \mu_e = e N_d \mu_e \quad [\because N_d \approx n_e]$$

$\therefore$ Number density of donor atoms or electron density is

$$N_d = \frac{\sigma}{e \mu_e}$$

Here, $\sigma = 5\Omega^{-1} \text{ cm}^{-1}$; $\mu_e = 3900 \text{ cm}^2/\text{Vs}$; $e = 1.6 \times 10^{-19} \text{ C}$

$$\therefore N_d = \frac{5}{3900 \times 1.6 \times 10^{-19}} = 8.01 \times 10^{15} \text{ cm}^{-3}$$

Example 3.9. Suppose a pure Si crystal has $5 \times 10^{28} \text{ atoms m}^{-3}$. It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$.

Solution. $1 \text{ ppm} = 1 \text{ part per million} = \frac{1}{10^6}$

$\therefore$ Number of pentavalent atoms doped in Si crystal

$$= \frac{5 \times 10^{28}}{10^6} = 5 \times 10^{22} \text{ m}^{-3}$$

As one pentavalent impurity atom donates 1 free electron to the crystal,

$\therefore$ Number of free electrons in the crystal is

$$n_e = 5 \times 10^{22} \text{ m}^{-3}$$

$$\therefore \text{Number of holes, } n_h = \frac{n_i^2}{n_e} = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}} = 4.5 \times 10^9 \text{ m}^{-3}$$

3.19 pn Junction

When a p -type semiconductor is suitably joined to n -type semiconductor, the contact surface is called **pn junction**.

Most semiconductor devices contain one or more *pn* junctions. The *pn* junction is of great importance because it is in effect, the *control element* for semiconductor devices. A thorough knowledge of the formation and properties of *pn* junction can enable the reader to understand the semiconductor devices.

Formation of *pn* junction. In actual practice, the characteristic properties of *pn* junction will not be apparent if a *p*-type block is just brought in contact with *n*-type block. In fact, *pn* junction is fabricated by special techniques. One common method of making *pn* junction is called *alloying*. In this method, a small block of indium (trivalent impurity) is placed on an *n*-type germanium slab as shown in Fig. 3.19 (i). The system is then heated to a temperature of about 500°C. The indium and some of the germanium melt to form a small puddle of molten germanium-indium mixture as shown in Fig. 3.19 (ii). The temperature is then lowered and puddle begins to solidify. Under proper conditions, the atoms of indium impurity will be suitably adjusted in the germanium slab to form a single crystal. The addition of indium overcomes the excess of electrons in the *n*-type germanium to such an extent that it creates a *p*-type region.

As the process goes on, the remaining molten mixture becomes increasingly rich in indium. When all germanium has been redeposited, the remaining material appears as indium button which is frozen on to the outer surface of the crystallised portion as shown in Fig. 3.19 (iii). This button serves as a suitable base for soldering on leads.

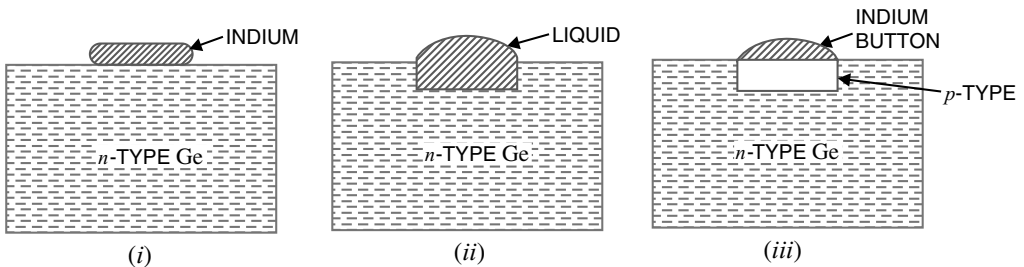


Fig. 3.19

3.20 Properties of *pn* Junction

At the instant of *pn*-junction formation, the free electrons near the junction in the *n* region begin to diffuse across the junction into the *p* region where they combine with holes near the junction. The result is that *n* region loses free electrons as they diffuse into the junction. This creates a layer of positive charges (pentavalent ions) near the junction. As the electrons move across the junction, the *p* region loses holes as the electrons and holes combine. The result is that there is a layer of negative charges (trivalent ions) near the junction. These two layers of positive and negative charges form the *depletion region* (or *depletion layer*). The term depletion is due to the fact that near the junction, the region is depleted (*i.e.* emptied) of *charge carriers* (free electrons and holes) due to diffusion across the junction. It may be noted that depletion layer is formed very quickly and is very thin compared to the *n* region and the *p* region. For clarity, the width of the depletion layer is shown exaggerated.

Once *pn* junction is formed and depletion layer created, the diffusion of free electrons stops. In other words, the depletion region acts as a barrier to the further movement of free electrons

3.23 Volt-Ampere Characteristics of pn Junction

Volt-ampere or V - I characteristic of a pn junction (also called a *crystal or semiconductor diode*) is the curve between voltage across the junction and the circuit current. Usually, voltage is taken along x -axis and current along y -axis. Fig. 3.25 shows the *circuit arrangement for determining the V - I characteristics of a pn junction. The characteristics can be studied under three heads, namely ; *zero external voltage, forward bias and reverse bias*.

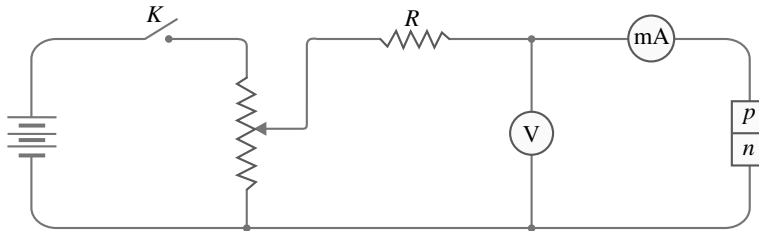


Fig. 3.25

(i) **Zero external voltage.** When the external voltage is zero, *i.e.* circuit is open at K , the potential barrier at the junction does not permit current flow. Therefore, the circuit current is zero as indicated by point O in Fig. 3.26.

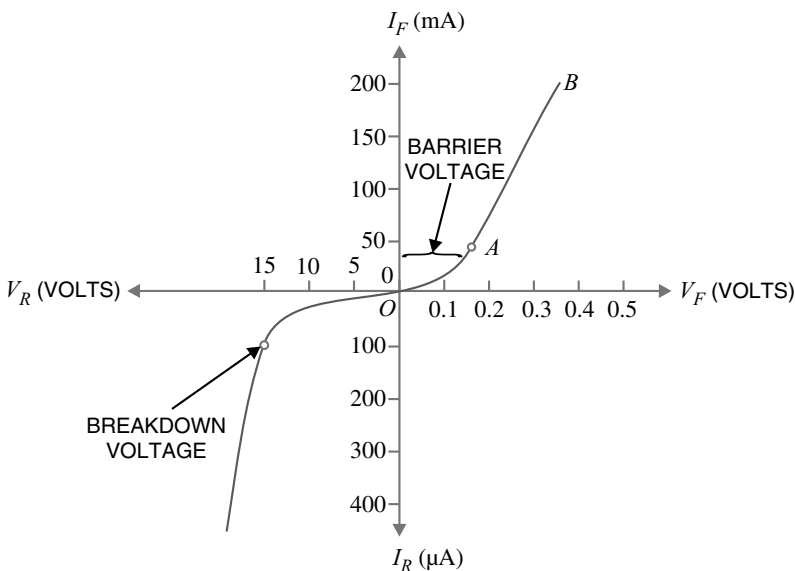


Fig. 3.26

(ii) **Forward bias.** With forward bias to the pn junction *i.e.* p -type connected to positive terminal and n -type connected to negative terminal, the potential barrier is reduced. At some forward voltage (0.7 V for Si and 0.3 V for Ge), the potential barrier is altogether eliminated and current starts flowing in the circuit. From now onwards, the current increases with the increase in forward voltage. Thus, a rising curve OB is obtained with forward bias as shown in Fig. 3.26. From the forward characteristic, it is seen that at first (*region OA*), the current

* R is the current limiting resistance. It prevents the forward current from exceeding the permitted value.

important when diode is used as a rectifier. In rectifier service, it has to be ensured that reverse voltage across the diode does not exceed its *PIV* during the negative half-cycle of input a.c. voltage. As a matter of fact, *PIV* consideration is generally the deciding factor in diode rectifier circuits. The peak inverse voltage may be between 10 V and 10 kV depending upon the type of diode.

(iii) Reverse current or leakage current. It is the current that flows through a reverse biased diode. This current is due to the minority carriers. Under normal operating voltages, the reverse current is quite small. Its value is extremely small ($< 1\mu\text{A}$) for silicon diodes but it is appreciable ($\approx 100\mu\text{A}$) for germanium diodes.

It may be noted that the reverse current is usually very small as compared with forward current. For example, the forward current for a typical diode might range upto 100 mA while the reverse current might be only a few μA —a ratio of many thousands between forward and reverse currents.

4.9 Semiconductor Diode Rectifiers

For reasons associated with economics of generation and transmission, the electric power available is usually an a.c. supply. The supply voltage varies sinusoidally and has a frequency of 50 Hz. It is used for lighting, heating and electric motors. But there are many applications (e.g. electronic circuits) where d.c. supply is needed. When such a d.c. supply is required, the mains a.c. supply is rectified by using semiconductor diodes. The following two rectifier circuits can be used :

(i) Half-wave rectifier

(ii) Full-wave rectifier

4.10 Half-Wave Rectifier

In half-wave rectification, the rectifier conducts current only during the positive half-cycles of input a.c. supply. The negative half-cycles of a.c. supply are suppressed *i.e.* during negative half-cycles, no current is conducted and hence no voltage appears across the load. Therefore, current always flows in one direction (*i.e.* d.c.) through the load though after every half-cycle.

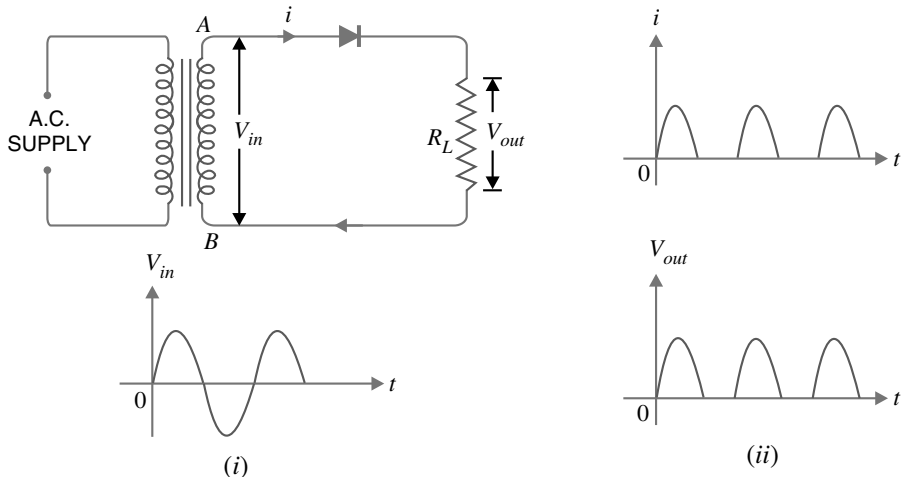


Fig. 4.26

Circuit details. Fig. 4.26 shows the circuit where a single semiconductor diode acts as a half-wave rectifier. The a.c. supply to be rectified is applied in series with the diode and load resistance R_L . Generally, a.c. supply is given through a transformer. The use of transformer permits two advantages. Firstly, it allows us to step-up or step-down the a.c. input voltage as the situation demands. Secondly, the transformer isolates the rectifier circuit from power line and thus reduces the risk of electric shock.

Operation. The a.c. voltage across the secondary winding AB changes polarities after every half-cycle. During the positive half-cycle of input a.c. voltage, end A becomes positive *w.r.t.* end B . This makes the diode forward biased and hence it conducts current. During the negative half-cycle, end A is negative *w.r.t.* end B . Under this condition, the diode is reverse biased and it conducts no current. Therefore, current flows through the diode during positive half-cycles of input a.c. voltage only ; it is blocked during the negative half-cycles [See Fig. 4.26 (ii)]. In this way, current flows through load R_L always in the same direction. Hence d.c. output is obtained across R_L . It may be noted that output across the load is pulsating d.c. These pulsations in the output are further smoothened with the help of *filter circuits* discussed later.

Disadvantages. The main disadvantages of a half-wave rectifier are :

(i) The pulsating current in the load contains alternating component whose basic frequency is equal to the supply frequency. Therefore, an elaborate filtering is required to produce steady direct current.

(ii) The a.c. supply delivers power only half the time. Therefore, the output is low.

4.11 Output Frequency of Half-Wave Rectifier

The output frequency of a half-wave rectifier is equal to the input frequency (50 Hz). Recall how a complete cycle is defined. A waveform has a complete cycle when it repeats the same wave pattern over a given time. Thus in Fig. 4.27 (i), the a.c. input voltage repeats the same wave pattern over $0^\circ - 360^\circ$, $360^\circ - 720^\circ$ and so on. In Fig. 4.27 (ii), the output waveform also repeats the same wave pattern over $0^\circ - 360^\circ$, $360^\circ - 720^\circ$ and so on. This means that when input a.c. completes one cycle, the output half-wave rectified wave also completes one cycle. In other words, the output frequency is equal to the input frequency *i.e.*

$$f_{out} = f_{in}$$

For example, if the input frequency of sine wave applied to a half-wave rectifier is 100 Hz, then frequency of the output wave will also be 100 Hz.

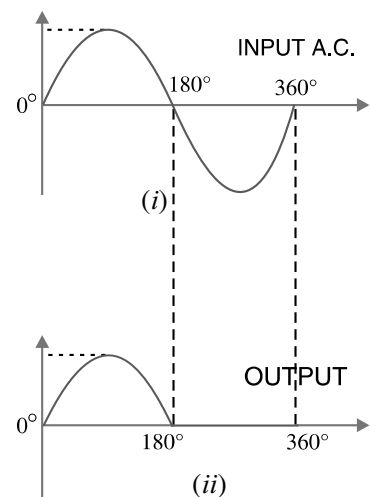


Fig. 4.27

4.12 Efficiency of Half-Wave Rectifier

The ratio of d.c. power output to the applied input a.c. power in a rectifier is known as rectifier efficiency *i.e.*

$$\text{Rectifier efficiency, } \eta = \frac{\text{D.C. power output}}{\text{Input a.c. power}}$$

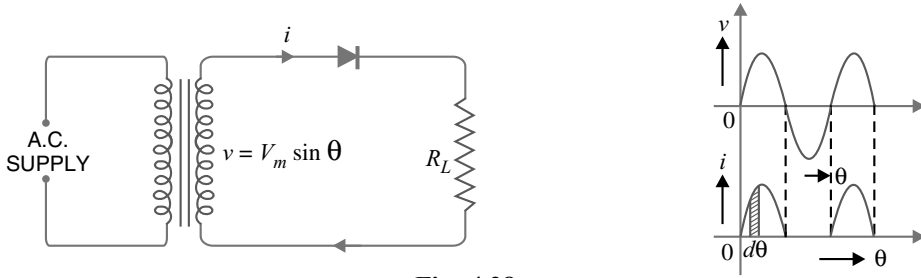


Fig. 4.28

Consider a half-wave rectifier shown in Fig. 4.28. Let $v = V_m \sin \theta$ be the alternating voltage that appears across the secondary winding. Let r_f and R_L be the diode resistance and load resistance respectively. The diode conducts during positive half-cycles of a.c. supply while no current conduction takes place during negative half-cycles.

d.c. power. The output current is pulsating direct current. Therefore, in order to find d.c. power, average current has to be found out.

$$\begin{aligned}
 *I_{av} = I_{dc} &= \frac{1}{2\pi} \int_0^\pi i \, d\theta = \frac{1}{2\pi} \int_0^\pi \frac{V_m \sin \theta}{r_f + R_L} \, d\theta \\
 &= \frac{V_m}{2\pi(r_f + R_L)} \int_0^\pi \sin \theta \, d\theta = \frac{V_m}{2\pi(r_f + R_L)} [-\cos \theta]_0^\pi \\
 &= \frac{V_m}{2\pi(r_f + R_L)} \times 2 = \frac{V_m}{(r_f + R_L)} \times \frac{1}{\pi} \\
 &= ** \frac{I_m}{\pi} \quad \left[\because I_m = \frac{V_m}{(r_f + R_L)} \right]
 \end{aligned}$$

$$\therefore \text{d.c. power, } P_{dc} = I_{dc}^2 \times R_L = \left(\frac{I_m}{\pi} \right)^2 \times R_L \quad \dots(i)$$

a.c. power input. The a.c. power input is given by ;

$$P_{ac} = I_{rms}^2 (r_f + R_L)$$

For a half-wave rectified wave, $I_{rms} = I_m/2$

$$\therefore P_{ac} = \left(\frac{I_m}{2} \right)^2 \times (r_f + R_L) \quad \dots(ii)$$

$$\begin{aligned}
 \therefore \text{Rectifier efficiency} &= \frac{\text{d.c. output power}}{\text{a.c. input power}} = \frac{(I_m / \pi)^2 \times R_L}{(I_m / 2)^2 (r_f + R_L)} \\
 &= \frac{0.406 R_L}{r_f + R_L} = \frac{0.406}{1 + \frac{r_f}{R_L}}
 \end{aligned}$$

$$* \text{ Average value} = \frac{\text{Area under the curve over a cycle}}{\text{Base}} = \frac{\int_0^\pi i \, d\theta}{2\pi}$$

** It may be remembered that the area of one-half cycle of a sinusoidal wave is twice the peak value. Thus in this case, peak value is I_m and, therefore, area of one-half cycle is $2 I_m$.

$$\therefore I_{av} = I_{dc} = \frac{2 I_m}{2\pi} = \frac{I_m}{\pi}$$

The efficiency will be maximum if r_f is negligible as compared to R_L .

∴ Max. rectifier efficiency = 40.6%

This shows that in half-wave rectification, a maximum of 40.6% of a.c. power is converted into d.c. power.

Example 4.23. The applied input a.c. power to a half-wave rectifier is 100 watts. The d.c. output power obtained is 40 watts.

(i) What is the rectification efficiency?

(ii) What happens to remaining 60 watts?

Solution.

(i) Rectification efficiency = $\frac{\text{d.c. output power}}{\text{a.c. input power}} = \frac{40}{100} = 0.4 = 40\%$

(ii) 40% efficiency of rectification does not mean that 60% of power is lost in the rectifier circuit. In fact, a semiconductor diode consumes little power due to its small internal resistance. The 100 W a.c. power is contained as 50 watts in positive half-cycles and 50 watts in negative half-cycles. The 50 watts in the negative half-cycles are not supplied at all. Only 50 watts in the positive half-cycles are converted into 40 watts.

∴ Power efficiency = $\frac{40}{50} \times 100 = 80\%$

Although 100 watts of a.c. power was supplied, the half-wave rectifier accepted only 50 watts and converted it into 40 watts d.c. power. Therefore, it is appropriate to say that efficiency of rectification is 40% and *not* 80% which is power efficiency.

Example 4.24. An a.c. supply of 230 V is applied to a half-wave rectifier circuit through a transformer of turn ratio 10 : 1. Find (i) the output d.c. voltage and (ii) the peak inverse voltage. Assume the diode to be ideal.

Solution.

Primary to secondary turns is

$$\frac{N_1}{N_2} = 10$$

R.M.S. primary voltage = 230 V

∴ Max. primary voltage is

$$\begin{aligned} V_{pm} &= (\sqrt{2}) \times \text{r.m.s. primary voltage} \\ &= (\sqrt{2}) \times 230 = 325.3 \text{ V} \end{aligned}$$

Max. secondary voltage is

$$V_{sm} = V_{pm} \times \frac{N_2}{N_1} = 325.3 \times \frac{1}{10} = 32.53 \text{ V}$$

(i)
$$I_{dc} = \frac{I_m}{\pi}$$

∴
$$V_{dc} = \frac{I_m}{\pi} \times R_L = \frac{V_{sm}}{\pi} = \frac{32.53}{\pi} = 10.36 \text{ V}$$

(ii) During the negative half-cycle of a.c. supply, the diode is reverse biased and hence conducts no current. Therefore, the maximum secondary voltage appears across the diode.

∴ Peak inverse voltage = 32.53 V

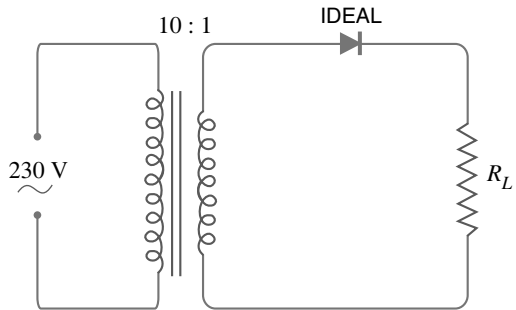


Fig. 4.29

Example 4.25. A crystal diode having internal resistance $r_f = 20\ \Omega$ is used for half-wave rectification. If the applied voltage $v = 50 \sin \omega t$ and load resistance $R_L = 800\ \Omega$, find :

- (i) I_m , I_{dc} , I_{rms} (ii) a.c. power input and d.c. power output
(iii) d.c. output voltage (iv) efficiency of rectification.

Solution.

$$v = 50 \sin \omega t$$

$$\therefore \text{Maximum voltage, } V_m = 50\text{ V}$$

$$(i) \quad I_m = \frac{V_m}{r_f + R_L} = \frac{50}{20 + 800} = 0.061\text{ A} = \mathbf{61\text{ mA}}$$

$$I_{dc} = I_m / \pi = 61 / \pi = \mathbf{19.4\text{ mA}}$$

$$I_{rms} = I_m / 2 = 61 / 2 = \mathbf{30.5\text{ mA}}$$

$$(ii) \quad \begin{aligned} \text{a.c. power input} &= (I_{rms})^2 \times (r_f + R_L) = \left(\frac{30.5}{1000} \right)^2 \times (20 + 800) \\ &= \mathbf{0.763\text{ watt}} \end{aligned}$$

$$\text{d.c. power output} = I_{dc}^2 \times R_L = \left(\frac{19.4}{1000} \right)^2 \times 800 = \mathbf{0.301\text{ watt}}$$

$$(iii) \quad \text{d.c. output voltage} = I_{dc} R_L = 19.4\text{ mA} \times 800\ \Omega = \mathbf{15.52\text{ volts}}$$

$$(iv) \quad \text{Efficiency of rectification} = \frac{0.301}{0.763} \times 100 = \mathbf{39.5\%}$$

Example 4.26. A half-wave rectifier is used to supply 50 V d.c. to a resistive load of $800\ \Omega$. The diode has a resistance of $25\ \Omega$. Calculate a.c. voltage required.

Solution.

$$\text{Output d.c. voltage, } V_{dc} = 50\text{ V}$$

$$\text{Diode resistance, } r_f = 25\ \Omega$$

$$\text{Load resistance, } R_L = 800\ \Omega$$

Let V_m be the maximum value of a.c. voltage required.

$$\begin{aligned} \therefore \quad V_{dc} &= I_{dc} \times R_L \\ &= \frac{I_m}{\pi} \times R_L = \frac{V_m}{\pi(r_f + R_L)} \times R_L \quad \left[\because I_m = \frac{V_m}{r_f + R_L} \right] \end{aligned}$$

$$\text{or} \quad 50 = \frac{V_m}{\pi(25 + 800)} \times 800$$

$$\therefore \quad V_m = \frac{\pi \times 825 \times 50}{800} = \mathbf{162\text{ V}}$$

Hence a.c. voltage of maximum value 162 V is required.

Example 4.27. Fig. 4.30 shows a circuit for charging a 12 V battery. If v_s is a sinusoid with 24 V peak value, find (i) the fraction of each cycle during which the diode conducts (ii) the peak value of the diode current and (iii) the maximum reverse-bias voltage that appears across the diode.

Solution. (i) The diode conducts when v_s exceeds 12 V.

$$\text{Now, } 24 \sin \theta = 12 \text{ or } \sin \theta = 1/2 \quad \therefore \theta = 30^\circ$$

$$\therefore \text{Conduction angle} = 180^\circ - 2 \times 30^\circ = 120^\circ$$

Therefore, during the positive half-cycle of v_s , the diode conducts for 120° (i.e. from 30° to 150°). However, during the negative half-cycle of v_s , the diode is reverse biased and does not conduct. This means that for 1 cycle ($= 360^\circ$) of v_s , the diode conducts for 120° or **one-third of 1 cycle of v_s** .

$$\begin{aligned} \text{(ii) Peak diode current} &= \frac{24 \text{ V} - 12 \text{ V}}{100 \Omega} \\ &= 0.12 \text{ A} \end{aligned}$$

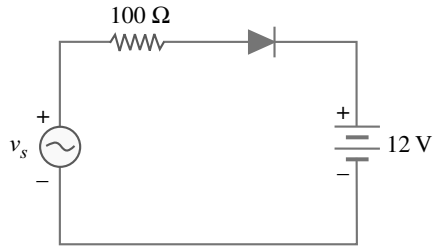


Fig. 4.30

(iii) During the negative half-cycle of v_s , the maximum reverse voltage across the diode occurs when v_s is at its negative peak. Therefore, maximum reverse voltage across the diode $= 24 \text{ V} + 12 \text{ V} = 36 \text{ V}$.

4.13 Full-Wave Rectifier

In full-wave rectification, current flows through the load in the same direction for both half-cycles of input a.c. voltage. This can be achieved with two diodes working alternately. For the positive half-cycle of input voltage, one diode supplies current to the load and for the negative half-cycle, the other diode does so ; current being always in the same direction through the load. Therefore, a full-wave rectifier utilises both half-cycles of input a.c. voltage to produce the d.c. output. The following two circuits are commonly used for full-wave rectification :

- (i) Centre-tap full-wave rectifier (ii) Full-wave bridge rectifier

4.14 Centre-Tap Full-Wave Rectifier

The circuit employs two diodes D_1 and D_2 as shown in Fig. 4.31. A centre tapped secondary winding AB is used with two diodes connected so that each uses one half-cycle of input a.c. voltage. In other words, diode D_1 utilises the a.c. voltage appearing across the upper half (OA) of secondary winding for rectification while diode D_2 uses the lower half winding OB .

Operation. During the positive half-cycle of secondary voltage, the end A of the secondary winding becomes positive and end B negative. This makes the diode D_1 forward biased and diode D_2 reverse biased. Therefore, diode D_1 conducts while diode D_2 does not. The conventional current flow is through diode D_1 , load resistor R_L and the upper half of secondary winding as shown by the dotted arrows. During the negative half-cycle, end A of the secondary winding becomes negative and end B positive. Therefore, diode D_2 conducts while diode D_1 does not. The conventional current flow is through diode D_2 , load R_L and lower half winding as shown by solid arrows. Referring to Fig. 4.31, it may be seen that current in the load R_L is *in the same direction* for both half-cycles of input a.c. voltage. Therefore, d.c. is obtained across the load R_L . Also, the polarities of the d.c. output across the load should be noted.

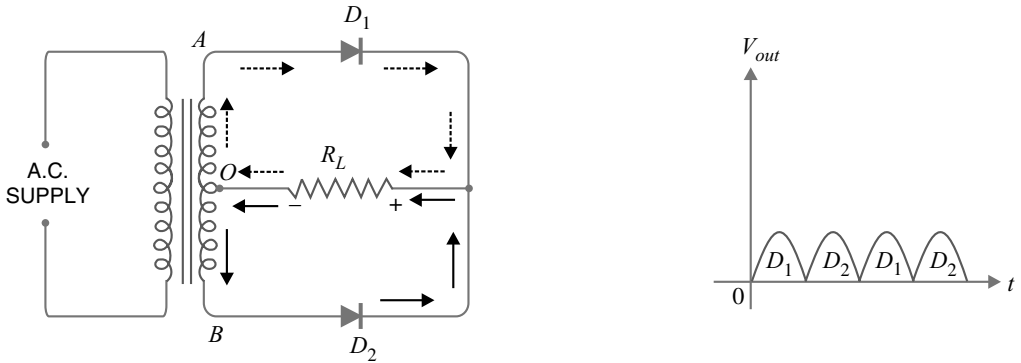


Fig. 4.31

Peak inverse voltage. Suppose V_m is the maximum voltage across the half secondary winding. Fig. 4.32 shows the circuit at the instant secondary voltage reaches its maximum value in the positive direction. At this instant, diode D_1 is conducting while diode D_2 is non-conducting. Therefore, whole of the secondary voltage appears across the non-conducting diode. Consequently, the peak inverse voltage is twice the maximum voltage across the half-secondary winding i.e.

$$PIV = 2V_m$$

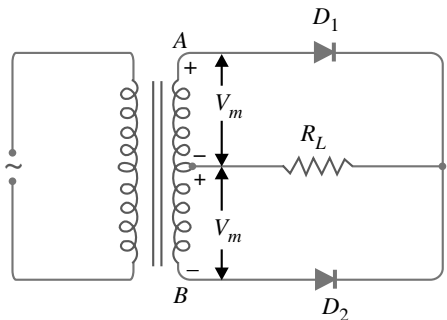


Fig. 4.32

Disadvantages

- (i) It is difficult to locate the centre tap on the secondary winding.
- (ii) The d.c. output is small as each diode utilises only one-half of the transformer secondary voltage.
- (iii) The diodes used must have high peak inverse voltage.

4.15 Full-Wave Bridge Rectifier

The need for a centre tapped power transformer is eliminated in the bridge rectifier. It contains four diodes D_1 , D_2 , D_3 and D_4 connected to form bridge as shown in Fig. 4.33. The a.c. supply to be rectified is applied to the diagonally opposite ends of the bridge through the transformer. Between other two ends of the bridge, the load resistance R_L is connected.

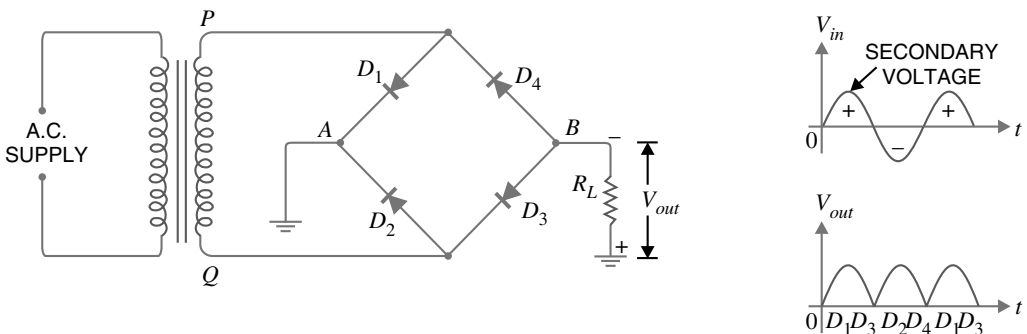


Fig. 4.33

Operation. During the positive half-cycle of secondary voltage, the end P of the secondary winding becomes positive and end Q negative. This makes diodes D_1 and D_3 forward biased while diodes D_2 and D_4 are reverse biased. Therefore, only diodes D_1 and D_3 conduct. These two diodes will be in series through the load R_L as shown in Fig. 4.34 (i). The conventional current flow is shown by dotted arrows. It may be seen that current flows from A to B through the load R_L .

During the negative half-cycle of secondary voltage, end P becomes negative and end Q positive. This makes diodes D_2 and D_4 forward biased whereas diodes D_1 and D_3 are reverse biased. Therefore, only diodes D_2 and D_4 conduct. These two diodes will be in series through the load R_L as shown in Fig. 4.34 (ii). The current flow is shown by the solid arrows. It may be seen that again current flows from A to B through the load *i.e.* in the same direction as for the positive half-cycle. Therefore, d.c. output is obtained across load R_L .

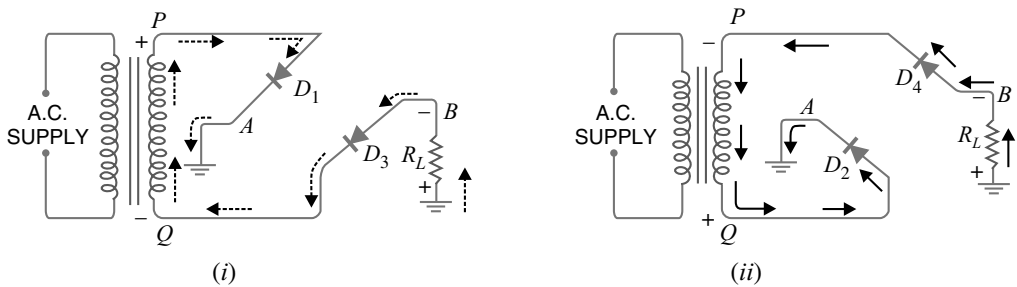


Fig. 4.34

Peak inverse voltage. The peak inverse voltage (PIV) of each diode is equal to the maximum secondary voltage of transformer. Suppose during positive half-cycle of input a.c., end P of secondary is positive and end Q negative. Under such conditions, diodes D_1 and D_3 are forward biased while diodes D_2 and D_4 are reverse biased. Since the diodes are considered ideal, diodes D_1 and D_3 can be replaced by wires as shown in Fig. 4.35 (i). This circuit is the same as shown in Fig. 4.35 (ii).

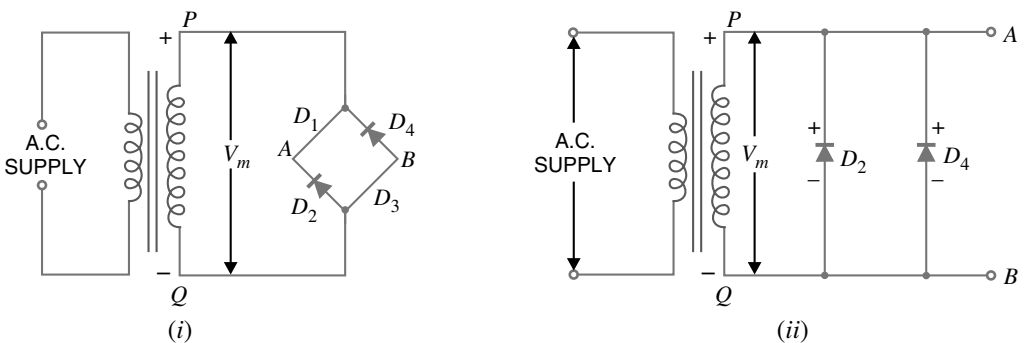


Fig. 4.35

Referring to Fig. 4.35 (ii), it is clear that two reverse biased diodes (*i.e.* D_2 and D_4) and the secondary of transformer are in parallel. Hence PIV of each diode (D_2 and D_4) is equal to the maximum voltage (V_m) across the secondary. Similarly, during the next half-cycle, D_2 and D_4 are forward biased while D_1 and D_3 will be reverse biased. It is easy to see that reverse voltage across D_1 and D_3 is equal to V_m .

Advantages

- (i) The need for centre-tapped transformer is eliminated.
- (ii) The output is twice that of the centre-tap circuit for the same secondary voltage.
- (iii) The *PIV* is one-half that of the centre-tap circuit (for same d.c. output).

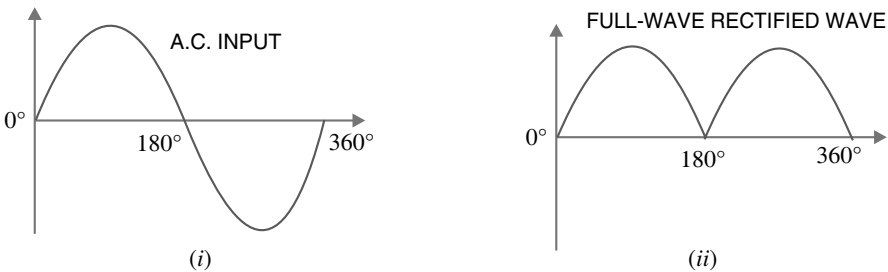
Disadvantages

- (i) It requires four diodes.
- (ii) As during each half-cycle of a.c. input two diodes that conduct are in series, therefore, voltage drop in the internal resistance of the rectifying unit will be twice as great as in the centre-tap circuit. This is objectionable when secondary voltage is small.

4.16 Output Frequency of Full-Wave Rectifier

The output frequency of a full-wave rectifier is double the input frequency. Remember that a wave has a complete cycle when it repeats the same pattern. In Fig. 4.36 (i), the input a.c. completes one cycle from $0^\circ - 360^\circ$. However, the full-wave rectified wave completes 2 cycles in this period [See Fig. 4.36 (ii)]. Therefore, output frequency is twice the input frequency *i.e.*

$$f_{out} = 2 f_{in}$$

**Fig. 4.36**

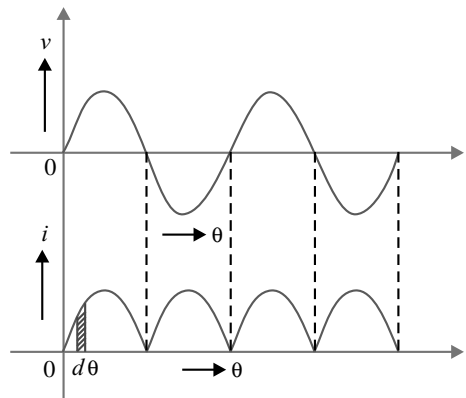
For example, if the input frequency to a full-wave rectifier is 100 Hz, then the output frequency will be 200 Hz.

4.17 Efficiency of Full-Wave Rectifier

Fig. 4.37 shows the process of full-wave rectification. Let $v = V_m \sin \theta$ be the a.c. voltage to be rectified. Let r_f and R_L be the diode resistance and load resistance respectively. Obviously, the rectifier will conduct current through the load in the same direction for both half-cycles of input a.c. voltage. The instantaneous current i is given by ;

$$i = \frac{v}{r_f + R_L} = \frac{V_m \sin \theta}{r_f + R_L}$$

d.c. output power. The output current is pulsating direct current. Therefore, in order to find the d.c. power, average current has to be found

**Fig. 4.37**

Solution. The lower the ripple factor of a power supply, the better it is.

For power supply A

$$\text{Ripple factor} = \frac{V_{ac(r.m.s.)}}{V_{dc}} \times 100 = \frac{0.5}{10} \times 100 = 5\%$$

For power supply B

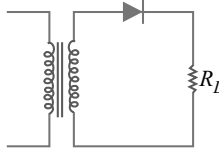
$$\text{Ripple factor} = \frac{V_{ac(r.m.s.)}}{V_{dc}} \times 100 = \frac{0.001}{25} \times 100 = 0.004\%$$

Clearly, power supply B is better.

4.21 Comparison of Rectifiers

Rectifier type : Half-wave

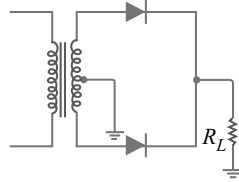
Schematic diagram :



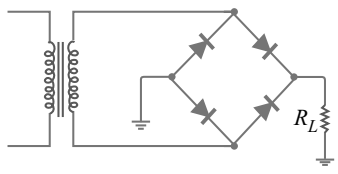
Typical output waveform :



Full-wave Centre-tap



Bridge Rectifier



S. No.	Particulars	Half-wave	Centre-tap	Bridge type
1.	No. of diodes	1	2	4
2.	Transformer necessary	no	yes	no
3.	Max. efficiency	40.6%	81.2%	81.2%
4.	Ripple factor	1.21	0.48	0.48
5.	Output frequency	f_{in}	$2 f_{in}$	$2 f_{in}$
6.	Peak inverse voltage	V_m	$2 V_m$	V_m

A comparison among the three rectifier circuits must be made very judiciously. Although bridge circuit has some disadvantages, it is the best circuit from the viewpoint of overall performance. When cost of the transformer is the main consideration in a rectifier assembly, we invariably use the bridge circuit. This is particularly true for large rectifiers which have a low-voltage and a high-current rating.

4.22 Filter Circuits

Generally, a rectifier is required to produce pure d.c. supply for using at various places in the electronic circuits. However, the output of a rectifier has pulsating *character *i.e.* it contains a.c. and d.c. components. The a.c. component is undesirable and must be kept away from the load. To do so, a *filter circuit* is used which removes (or *filters out*) the a.c. component and allows only the d.c. component to reach the load.

A filter circuit is a device which removes the a.c. component of rectifier output but allows the d.c. component to reach the load.

Obviously, a filter circuit should be installed between the rectifier and the load as shown in

* If such a d.c. is applied in an electronic circuit, it will produce a hum.

Fig. 4.49. A filter circuit is generally a combination of inductors (L) and capacitors (C). The filtering action of L and C depends upon the basic electrical principles. A capacitor passes a.c. readily but does not *pass d.c. at all. On the other hand, an inductor †opposes a.c. but allows d.c. to pass through it. It then becomes clear that suitable network of L and C can effectively remove the a.c. component, allowing the d.c. component to reach the load.

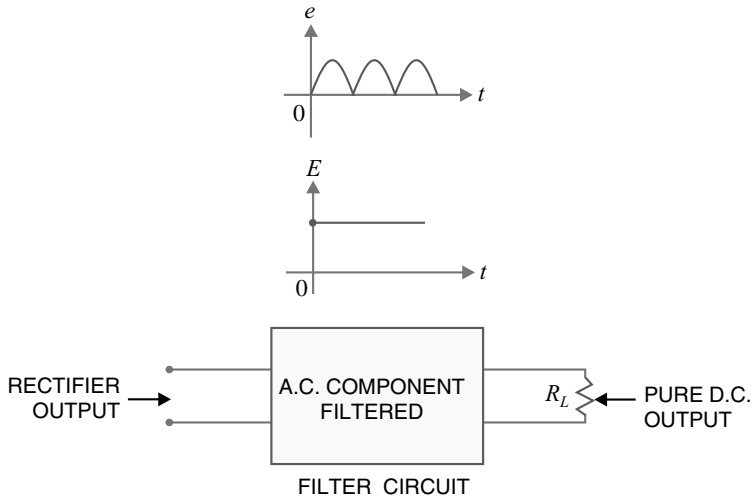


Fig. 4.49

4.23 Types of Filter Circuits

The most commonly used filter circuits are *capacitor filter*, *choke input filter* and *capacitor input filter or π -filter*. We shall discuss these filters in turn.

(i) **Capacitor filter.** Fig. 4.50 (ii) shows a typical capacitor filter circuit. It consists of a capacitor C placed across the rectifier output in parallel with load R_L . The pulsating direct voltage of the rectifier is applied across the capacitor. As the rectifier voltage increases, it charges the capacitor and also supplies current to the load. At the end of quarter cycle [Point A in Fig. 4.50 (iii)], the capacitor is charged to the peak value V_m of the rectifier voltage. Now, the rectifier voltage starts to decrease. As this occurs, the capacitor discharges through the load and voltage across it (*i.e.* across parallel combination of R - C) decreases as shown by the line AB in Fig. 4.50 (iii). The voltage across load will decrease only slightly because immediately the next voltage peak comes and recharges the capacitor. This process is repeated again and again and the output voltage waveform becomes $ABCDEFGH$. It may be seen that very little ripple is left in the output. Moreover, output voltage is higher as it remains substantially near the peak value of rectifier output voltage.

* A capacitor offers infinite reactance to d.c. For d.c., $f = 0$.

$$\therefore X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 0 \times C} = \infty$$

Hence, a capacitor does not allow d.c. to pass through it.

† We know $X_L = 2\pi fL$. For d.c., $f = 0$ and, therefore, $X_L = 0$. Hence inductor passes d.c. quite readily. For a.c., it offers opposition and drops a part of it.

$$\therefore V_{dc} = V_{p(in)} \left(1 - \frac{1}{2f R_L C} \right) = 14.9 (1 - 0.038) = 14.3 \text{ V}$$

Example 4.39. The choke of Fig. 4.54 has a d.c. resistance of 25Ω . What is the d.c. voltage if the full-wave signal into the choke has a peak value of 25.7 V ?

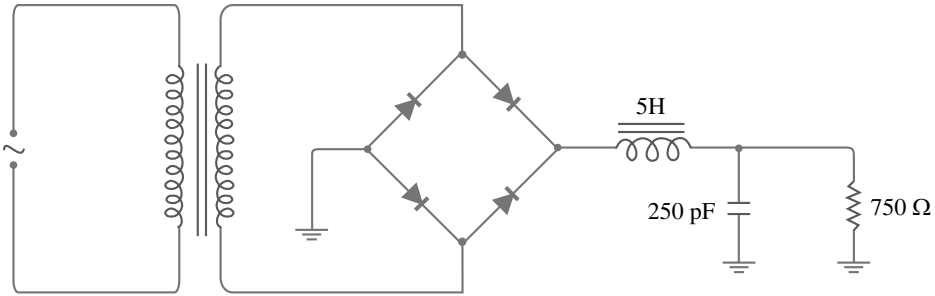


Fig. 4.54

Solution. The output of a full-wave rectifier has a d.c. component and an a.c. component. Due to the presence of a.c. component, the rectifier output has a pulsating character as shown in Fig. 4.55. The maximum value of the pulsating output is V_m and d.c. component is $V'_{dc} = 2V_m/\pi$.

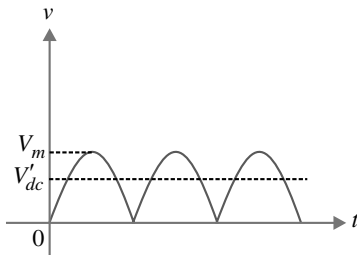


Fig. 4.55

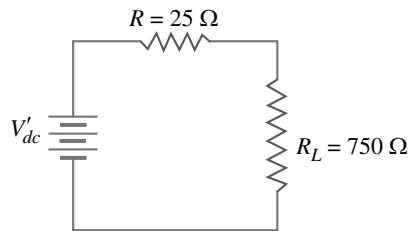


Fig. 4.56

For d.c. component V'_{dc} , the choke resistance is in series with the load as shown in Fig. 4.56.

$$\therefore \text{Voltage across load, } V_{dc} = \frac{V'_{dc}}{R + R_L} \times R_L$$

$$\text{In our example, } V'_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 25.7}{\pi} = 16.4 \text{ V}$$

$$\therefore \text{Voltage across load, } V_{dc} = \frac{V'_{dc}}{R + R_L} \times R_L = \frac{16.4}{25 + 750} \times 750 = 15.9 \text{ V}$$

The voltage across the load is 15.9 V d.c. *plus* a small ripple.

4.24 Voltage Multipliers

With a diode, we can build a rectifier to produce a d.c. voltage that is nearly equal to the peak value of input a.c. voltage. We can also use diodes and capacitors to build a circuit that will provide a *d.c. output* that is multiple of the *peak input a.c. voltage*. Such a circuit

is called a voltage multiplier. For example, a *voltage doubler* will provide a d.c. output that is *twice* the peak input a.c. voltage, a *voltage tripler* will provide a d.c. output that is *three times* the peak input a.c. voltage and so on.

While voltage multipliers provide d.c. output that is much greater than the peak input a.c. voltage, there is no power amplification and law of conservation of energy holds good. When a voltage multiplier *increases* the peak input voltage by a factor n , the peak input current is *decreased* by approximately the same factor. Thus the actual power output from a voltage multiplier will *never* be greater than the input power. In fact, there are losses in the circuit (e.g. in diodes, capacitors *etc.*) so that the output power will actually be *less than* the input power.

4.25 Half-Wave Voltage Doubler

A half-wave voltage doubler consists of two diodes and two capacitors connected in a manner as shown in Fig. 4.57. It will be shown that if the peak input a.c. voltage is $V_{S(pk)}$, the d.c. output voltage will be $2V_{S(pk)}$ provided the diodes are ideal (this assumption is fairly reasonable). The basic idea in a voltage multiplier is to charge each capacitor to the peak input a.c. voltage and to arrange the capacitors so that their stored voltages will add.

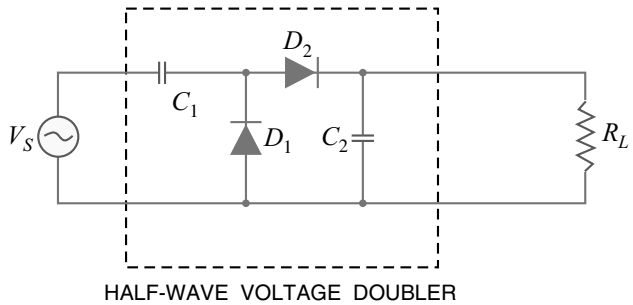


Fig. 4.57

Circuit action. We now discuss the working of a half-wave voltage doubler.

(i) During the negative half-cycle of a.c. input voltage [See Fig. 4.58 (i)], diode D_1 is forward biased and diode D_2 is reverse biased [See Fig. 4.58 (i)]. Therefore, diode D_1 can be represented by a *short* and diode D_2 as an *open*. The equivalent circuit then becomes as shown in Fig. 4.58 (ii).

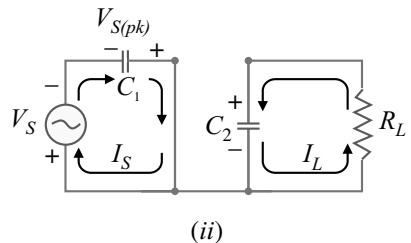
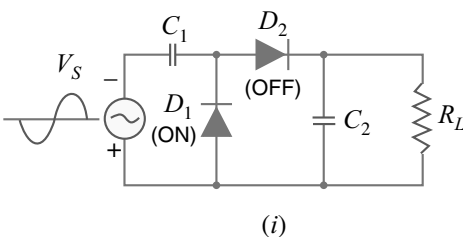


Fig. 4.58

As you can see [See Fig. 4.58 (ii)], C_1 will charge until voltage across it becomes equal to peak value of source voltage $[V_{S(pk)}]$. At the same time, C_2 will be in the process of discharging through the load R_L (The source of this charge on C_2 will be explained in a moment). *Note that in all figures electron flow is shown.*

regardless of the variations in the input voltage or load. In order to ensure this, a voltage stabilising device, called voltage stabiliser is used. Several stabilising circuits have been designed but only *zener diode* as a voltage stabiliser will be discussed.

4.27 Zener Diode

It has already been discussed that when the reverse bias on a crystal diode is increased, a critical voltage, called *breakdown voltage* is reached where the reverse current increases sharply to a high value. The breakdown region is the knee of the reverse characteristic as shown in Fig. 4.61. The satisfactory explanation of this breakdown of the junction was first given by the American scientist C. Zener. Therefore, the breakdown voltage is sometimes called *zener voltage* and the sudden increase in current is known as *zener current*.

The breakdown or zener voltage depends upon the amount of doping. If the diode is heavily doped, depletion layer will be thin and consequently the breakdown of the junction will occur at a lower reverse voltage. On the other hand, a lightly doped diode has a higher breakdown voltage. When an ordinary crystal diode is properly doped so that it has a sharp breakdown voltage, it is called a zener diode.

A properly doped crystal diode which has a sharp breakdown voltage is known as a zener diode.

Fig. 4.62 shows the symbol of a zener diode. It may be seen that it is just like an ordinary diode except that the bar is turned into z-shape. The following points may be noted about the zener diode :

- (i) A zener diode is like an ordinary diode except that it is properly doped so as to have a sharp breakdown voltage.
- (ii) A zener diode is always reverse connected *i.e.* it is always reverse biased.
- (iii) A zener diode has sharp breakdown voltage, called zener voltage V_Z .
- (iv) When forward biased, its characteristics are just those of ordinary diode.
- (v) The zener diode is not immediately burnt just because it has entered the *breakdown region. As long as the external circuit connected to the diode limits the diode current to less than *burn out* value, the diode will not burn out.

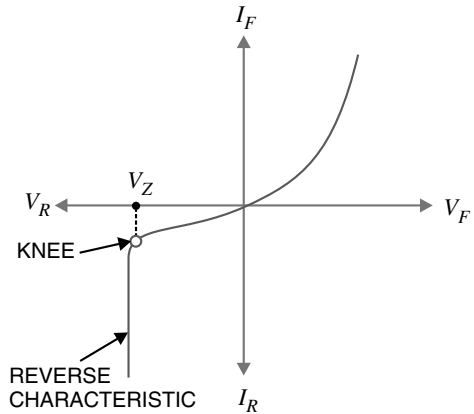


Fig. 4.61



Fig. 4.62

4.28 Reverse Breakdown of Diodes

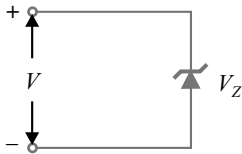
There is a limit to how much reverse voltage a diode can withstand before it is destroyed due to excessive heat. There are two types of diode breakdown due to reverse bias *viz.*

(i) Avalanche breakdown (ii) Zener breakdown.

(i) **Avalanche breakdown.** The avalanche breakdown occurs at high reverse voltages.

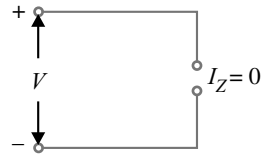
* The current is limited only by both external resistance and the power dissipation of zener diode.

(ii) “OFF” state. When the reverse voltage across the zener diode is less than V_Z but greater than 0 V, the zener diode is in the “OFF” state. Under such conditions, the zener diode can be represented by an open-circuit as shown in Fig. 4.64 (ii).



(i)

$$V_Z > V > 0$$



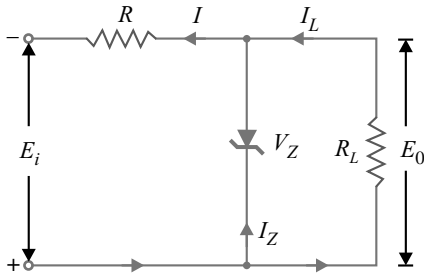
(ii)

EQUIVALENT CIRCUIT OF ZENER FOR “OFF” STATE

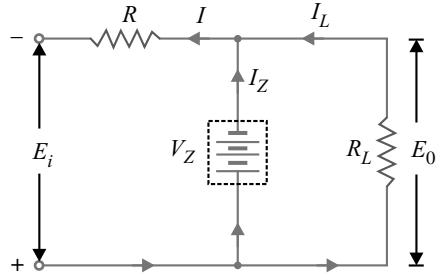
Fig. 4.64

4.30 Zener Diode as Voltage Stabiliser

A zener diode can be used as a voltage regulator to provide a constant voltage from a source whose voltage may vary over sufficient range. The circuit arrangement is shown in Fig. 4.65 (i). The zener diode of zener voltage V_Z is reverse connected across the load R_L across which constant output is desired. The series resistance R absorbs the output voltage fluctuations so as to maintain constant voltage across the load. It may be noted that the zener will maintain a constant voltage $V_Z (= E_0)$ across the load so long as the input voltage does not fall below V_Z .



(i)



(ii)

Fig. 4.65

When the circuit is properly designed, the load voltage E_0 remains essentially constant (equal to V_Z) even though the input voltage E_i and load resistance R_L may vary over a wide range.

(i) Suppose the input voltage increases. Since the zener is in the breakdown region, the zener diode is equivalent to a battery V_Z as shown in Fig. 4.65 (ii). It is clear that output voltage remains constant at $V_Z (= E_0)$. The excess voltage is dropped across the series resistance R . This will cause an increase in the value of total current I . The zener will conduct the increase of current in I while the load current remains constant. Hence, output voltage E_0 remains constant irrespective of the changes in the input voltage E_i .

(ii) Now suppose that input voltage is constant but the load resistance R_L decreases. This will cause an increase in load current. The extra current cannot come from the source because drop in R (and hence source current I) will not change as the zener is within its regulating range. The additional load current will come from a decrease in zener current I_Z . Consequently, the output voltage stays at constant value.

$$\text{Voltage drop across } R = E_i - E_0$$

$$\text{Current through } R, I = I_Z + I_L$$

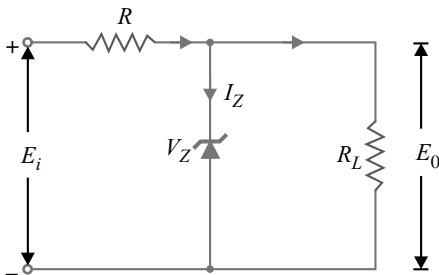
$$\text{Applying Ohm's law, we have, } R = \frac{E_i - E_0}{I_Z + I_L}$$

4.31 Solving Zener Diode Circuits

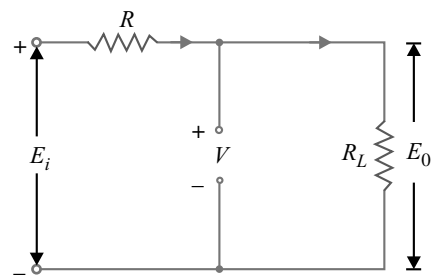
The analysis of zener diode circuits is quite similar to that applied to the analysis of semiconductor diodes. The first step is to determine the state of zener diode *i.e.* whether the zener is in the “on” state or “off” state. Next, the zener is replaced by its appropriate model. Finally, the unknown quantities are determined from the resulting circuit.

1. E_i and R_L fixed. This is the simplest case and is shown in Fig. 4.66 (i). Here the applied voltage E_i as well as load R_L is fixed. The first step is to find the state of zener diode. This can be determined by removing the zener from the circuit and calculating the voltage V across the resulting open-circuit as shown in Fig. 4.66 (ii).

$$V = E_0 = \frac{R_L E_i}{R + R_L}$$



(i)



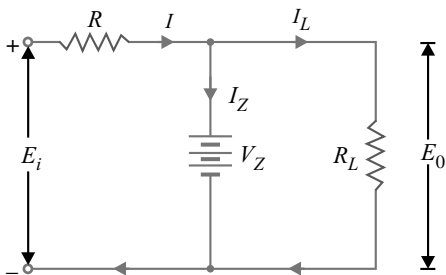
(ii)

Fig. 4.66

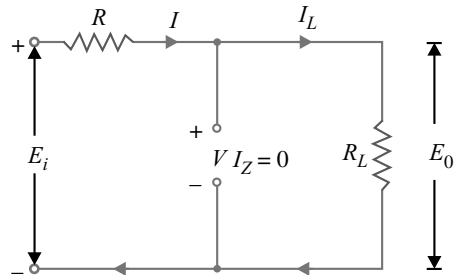
If $V \geq V_Z$, the zener diode is in the “on” state and its equivalent model can be substituted as shown in Fig. 4.67 (i). If $V < V_Z$, the diode is in the “off” state as shown in Fig. 4.67 (ii).

(i) On state. Referring to circuit shown in Fig. 4.67 (i),

$$E_0 = V_Z$$



(i)



(ii)

Fig. 4.67

$$I_Z = I - I_L \text{ where } I_L = \frac{E_0}{R_L} \text{ and } I = \frac{E_i - E_0}{R}$$

5.1 Zener Diode

A zener diode is a special type of diode that is designed to operate in the reverse breakdown region. An ordinary diode operated in this region will usually be destroyed due to excessive current. This is not the case for the zener diode.

A zener diode is heavily doped to reduce the reverse breakdown voltage. This causes a very thin depletion layer. As a result, a zener diode has a sharp reverse breakdown voltage V_Z . This is clear from the reverse characteristic of zener diode shown in Fig. 5.1. Note that the reverse characteristic drops in an almost vertical manner at reverse voltage V_Z . As the curve reveals, two things happen when V_Z is reached :

- (i) The diode current increases rapidly.
- (ii) The reverse voltage V_Z across the diode remains almost constant.

In other words, *the zener diode operated in this region will have a relatively constant voltage across it, regardless of the value of current through the device.* This permits the zener diode to be used as a *voltage regulator*. For detailed discussion on zener diode, the reader may refer to chapter 4 of this book.

5.2 Light-Emitting Diode (LED)

A light-emitting diode (LED) is a diode that gives off visible light when forward biased.

Light-emitting diodes are not made from silicon or germanium but are made by using elements like gallium, phosphorus and arsenic. By varying the quantities of these elements, it is possible to produce light of different wavelengths with colours that include red, green, yellow and blue. For example, when a LED is manufactured using gallium arsenide, it will produce a red light. If the LED is made with gallium phosphide, it will produce a green light.

Theory. When light-emitting diode (LED) is forward biased as shown in Fig. 5.2 (i), the electrons from the n -type material cross the pn junction and recombine with holes in the p -type material. Recall that these free electrons are in the conduction band and at a higher energy level than the holes in the valence band. When recombination takes place, the recombining electrons release energy in the form of heat and light. In germanium and silicon diodes, almost the entire energy is given up in the form of heat and emitted visible light is small. Since germanium and silicon are opaque materials, the emitted light is blocked *i.e.* it is not transmitted to the surroundings. However, in materials like **gallium arsenide** (GaAs) and **gallium phosphide** (GaP), *the number of photons of light energy is sufficient to produce quite intense visible light.* Since these materials are semi-transparent, some of the visible light escapes to the surroundings. LEDs have replaced incandescent lamps in many applications because of their low voltage, long life and fast on-off switching.

Fig. 5.2 (ii) shows the schematic symbol for a LED. The arrows are shown as pointing away from the diode, indicating that light is being emitted by the device when forward biased. Although LEDs are available in several colours (red, green, yellow and orange are the most common), the schematic symbol is the same for all LEDs. There is nothing in the symbol to indicate the colour of a particular LED. Fig. 5.3 shows the graph between radiated light and the forward current of the LED. It is clear from the graph that the intensity of radiated light is directly proportional to the forward current of LED.

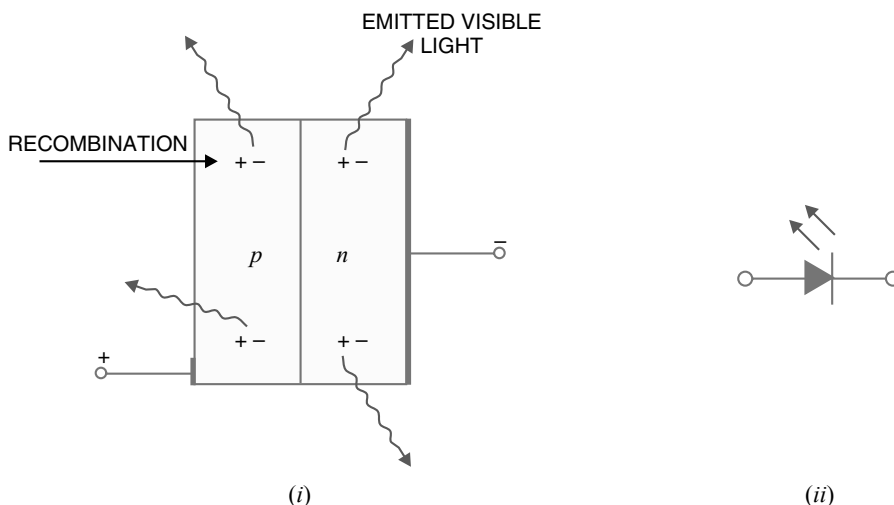


Fig. 5.2

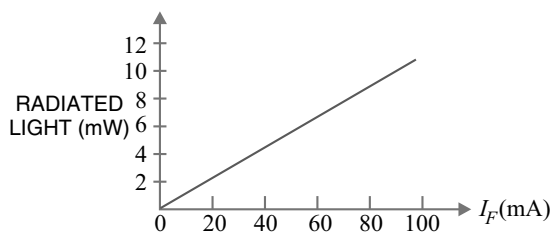


Fig. 5.3

5.3 LED Voltage and Current

The forward voltage ratings of most LEDs is from 1 V to 3 V and forward current ratings range from 20 mA to 100 mA. In order that current through the LED does not exceed the safe value, a resistor R_S is connected in series with it as shown in Fig. 5.4. The input voltage is V_S and the voltage across LED is V_D .

$$\therefore \text{Voltage across } R_S = V_S - V_D$$

$$\therefore \text{Circuit current, } I_F = \frac{V_S - V_D}{R_S}$$

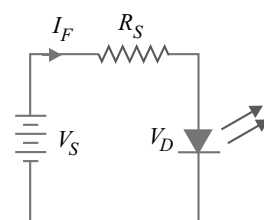


Fig. 5.4

The brightness of a LED is determined by the value of forward current (I_F) of the LED. The more the forward current, the brighter the light is. Therefore, we always keep the value of series **current-limiting resistor** R_S as small as possible to ensure that the LED is as bright as possible.

Note. Note that the forward voltage drop across a LED (2 V to 3 V) is considerably greater than that across a silicon or germanium diode (0.7 V for Si diode and 0.3 V for Ge diode) depending upon the forward current. However, the maximum permissible reverse voltage across a LED (5 V to 10 V) is relatively small.

Example 5.1. What value of series resistor is required to limit the current through a LED to 20 mA with a forward voltage drop of 1.6 V when connected to a 10 V supply?

Solution.

$$\text{Series resistor, } R_S = \frac{V_S - V_D}{I_F}$$

Here, $V_S = 10 \text{ V}$; $V_D = 1.6 \text{ V}$; $I_F = 20 \text{ mA} = 20 \times 10^{-3} \text{ A}$

$$\therefore R_S = \frac{10 - 1.6}{20 \times 10^{-3}} = 420 \, \Omega$$

Note that resistor R_S is also called *current-limiting resistor*.

Example 5.2. What is current through the LED in the circuit shown in Fig. 5.5? Assume that voltage drop across the LED is 2 V.

Solution.

$$\text{Current through LED, } I_F = \frac{V_S - V_D}{R_S}$$

Here, $V_S = 15 \text{ V}$; $V_D = 2 \text{ V}$; $R_S = 2.2 \text{ k}\Omega = 2.2 \times 10^3 \, \Omega$

$$\therefore I_F = \frac{15 - 2}{2.2 \times 10^3} = 5.91 \times 10^{-3} \text{ A} = 5.91 \text{ mA}$$

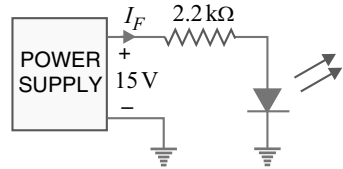


Fig. 5.5

Example 5.3. The driving circuit shown in Fig. 5.6 has a peak output voltage of 8 V. The LED has ratings of $V_D = 1.8 \text{ V}$ to 2.0 V and $I_F(\text{max.}) = 16 \text{ mA}$. What value of current-limiting resistor R_S is needed in the circuit ?

Solution.

$$\text{Forward current, } I_F = \frac{V_S - V_D}{R_S}$$

For maximum brightness of the LED, the forward current I_F should be as large as possible. It is given that $I_F(\text{max.}) = 16 \text{ mA}$ and $V_{S(\text{peak})} = 8 \text{ V}$. Therefore, we should use the minimum value of forward voltage drop of LED i.e. 1.8 V.

$$\therefore 16 \text{ mA} = \frac{8 \text{ V} - 1.8 \text{ V}}{R_S}$$

$$\text{or } R_S = \frac{8 \text{ V} - 1.8 \text{ V}}{16 \text{ mA}} = 387.5 \, \Omega$$

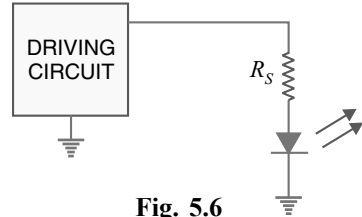


Fig. 5.6

Example 5.4. Bipolar junction transistor (BJT) is often used to switch on the LED. Fig. 5.7 shows a LED driving circuit designed to illuminate the LED when the transistor is turned on (saturated) by a 5V battery at the base. Find the values of R_B and R_C that should be used if the LED is to be illuminated by 20 mA of forward current. Assume that the forward voltage drop across the LED is 2.5 V and that the silicon transistor has $\beta = 50$.

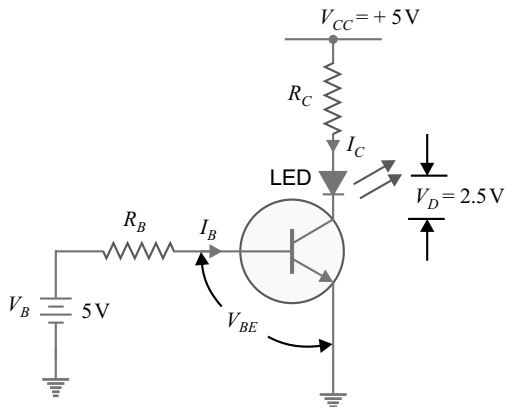


Fig. 5.7

Solution.

When the transistor is saturated, the value of I_C is given by ;

$$I_C = \frac{V_{CC} - V_D}{R_C} = \frac{5\text{ V} - 2.5\text{ V}}{R_C} = \frac{2.5\text{ V}}{R_C}$$

$$\therefore R_C = \frac{2.5\text{ V}}{I_C} = \frac{2.5\text{ V}}{20\text{ mA}} = 125\ \Omega$$

Now, Base current, $I_B = I_C / \beta = 20\text{ mA} / 50 = 0.4\text{ mA}$

$$\text{Also, } R_B = \frac{5 - V_{BE}}{I_B} = \frac{(5 - 0.7)\text{ V}}{0.4\text{ mA}} = 10.75\text{ k}\Omega$$

Note that the resistor R_B limits the transistor base current while the resistor R_C limits the LED current.

5.4 Advantages of LED

The light-emitting diode (LED) is a solid-state light source. LEDs have replaced incandescent lamps in many applications because they have the following advantages :

- (i) Low voltage
- (ii) Longer life (more than 20 years)
- (iii) Fast on-off switching (of the order of nanoseconds)

The principal disadvantage of the LED is that it draws considerable amount of current from the low-power circuits in which it is normally used. For this reason, LEDs are no longer used in such low-power devices as calculators and watches.

Protecting LED against reverse bias. The LEDs have low reverse voltage ratings. For example, a typical LED may have a maximum reverse voltage rating of 3 V. This means that if a reverse voltage greater than 3 V is applied to the LED, the LED may be destroyed. Therefore, one must be careful not to use LEDs with a high level of reverse bias. One way to protect a LED is to connect a rectifier diode in parallel with LED as shown in Fig. 5.8. If reverse voltage greater than the reverse voltage rating of LED is accidentally applied, the rectifier diode will be turned on. This protects the LED from damage.

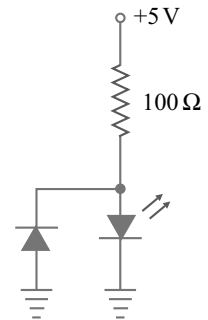


Fig. 5.8

5.5 Multicolour LEDs

A LED that emits one colour when forward biased and another colour when reverse biased is called a multicolour LED.

One commonly used schematic symbol for these LEDs is shown in Fig. 5.9. Multicolour LEDs actually contain two *pn* junctions that are connected in *reverse-parallel* i.e. they are in parallel with anode of one being connected to the cathode of the other. If positive potential is applied to the top terminal as shown in Fig. 5.9 (i), the *pn* junction on the left will light. Note that the device current passes through the left *pn* junction. If the polarity of the voltage source is reversed as shown in Fig. 5.9 (ii), the *pn* junction on the right will light. Note that the direction of device current has reversed and is now passing through the right *pn* junction.

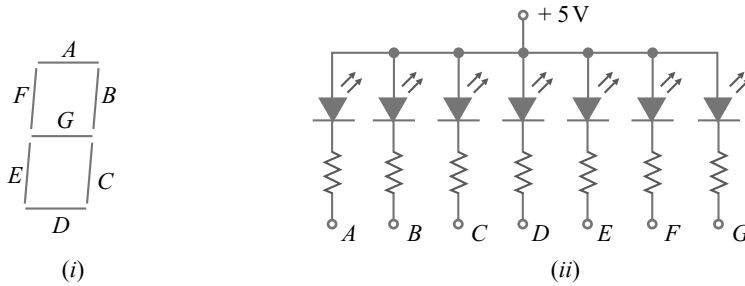


Fig. 5.11

Fig. 5.11 (ii) shows the schematic diagram of seven-segment display. External series resistors are included to limit currents to safe levels. Note that the anodes of all seven LEDs are connected to a common positive voltage source of +5 V. This arrangement is known as *common-anode type*. In order to light a particular LED, say *A*, we ground the point *A* in Fig. 5.11 (ii). It forward biases the LED *A* which will be lit.

5.7 Photo-diode

A **photo-diode** is a reverse-biased silicon or germanium *pn* junction in which reverse current increases when the junction is exposed to light.

The reverse current in a photo-diode is directly proportional to the intensity of light falling on its *pn* junction. This means that greater the intensity of light falling on the *pn* junction of photo-diode, the greater will be the reverse current.

Principle. When a rectifier diode is reverse biased, it has a very small reverse leakage current. The same is true for a photo-diode. The reverse current is produced by thermally generated electron-hole pairs which are swept across the junction by the electric field created by the reverse voltage. In a rectifier diode, the reverse current increases with temperature due to an increase in the number of electron-hole pairs. A photo-diode differs from a rectifier diode in that when its *pn* junction is exposed to light, the reverse current increases with the increase in light intensity and vice-versa. This is explained as follows. When light (photons) falls on the *pn* junction, the energy is imparted by the photons to the atoms in the junction. This will create more free electrons (and more holes). These additional free electrons will increase the reverse current. As the intensity of light incident on the *pn* junction increases, the reverse current also increases. In other words, as the incident light intensity increases, the resistance of the device (photo-diode) decreases.

Photo-diode package. Fig. 5.12 (i) shows a typical photo-diode package. It consists of a *pn* junction mounted on an insulated substrate and sealed inside a metal case. A glass window is mounted on top of the case to allow light to enter and strike the *pn* junction. The two leads extending from the case are labelled anode and cathode. The cathode is typically identified by a tab extending from the side of the case.

* Also available is the *common-cathode type* where all cathodes are connected together.

** This is true only if the light energy is applied at the junction. If it is applied to the crystal at some distance from the junction, the free electrons and holes will recombine (thus neutralising each other) before they can join the flow of reverse current.

*** It is for this reason that semiconductor devices such as diodes and transistors are usually enclosed in opaque case to protect them from light. Those diodes or transistors which are used for light-detecting, on the other hand, must be encased in transparent plastic or glass so that light may fall on them.

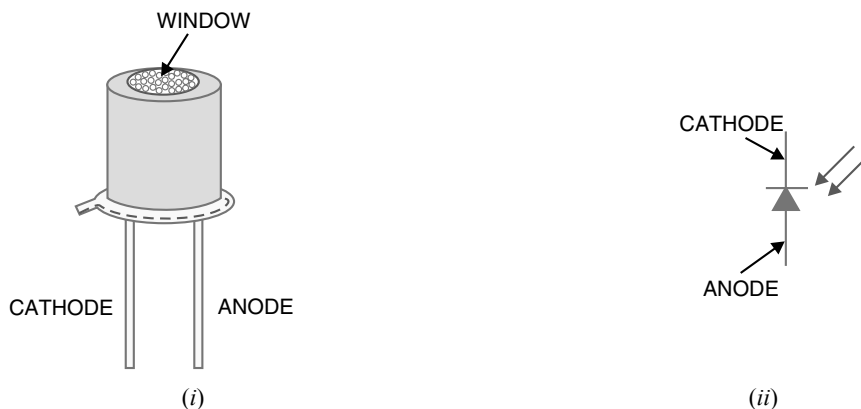


Fig. 5.12

Fig. 5.12 (ii) shows the schematic symbol of a photo-diode. The inward arrows represent the incoming light.

5.8 Photo-diode Operation

Fig. 5.13 shows the basic photo-diode circuit. The circuit has reverse-biased photo-diode, resistor R and d.c. supply. The operation of the photo-diode is as under :

(i) When no light is incident on the pn junction of photo-diode, the reverse current I_R is extremely small. This is called **dark current**.

The resistance of photo-diode with no incident light is called **dark resistance** (R_R).

$$\text{Dark resistance of photo-diode, } R_R = \frac{V_R}{\text{Dark current}}$$

(ii) When light is incident on the pn junction of the photo-diode, there is a transfer of energy from the incident light (photons) to the atoms in the junction. This will create more free electrons (and more holes). These additional free electrons will increase the reverse current.

(iii) As the intensity of light increases, the reverse current I_R goes on increasing till it becomes maximum. This is called **saturation current**.

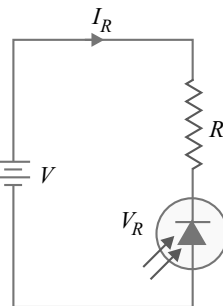


Fig. 5.13

5.9 Characteristics of Photo-diode

There are two important characteristics of photo-diode.

(i) **Reverse current-Illumination curve.** Fig. 5.14 shows the graph between reverse current (I_R) and illumination (E) of a photo-diode. The reverse current is shown on the vertical axis and is measured in μA . The illumination is indicated on the horizontal axis and is measured in mW/cm^2 . Note that graph is a straight line passing through the origin.

$$\therefore I_R = mE$$

where

$$m = \text{Slope of the straight line}$$

The quantity m is called the *sensitivity* of the photo-diode.

Transistors

Learning Objectives

- Develop an understanding of bipolar junction transistor (BJT) and its operation.
- Learn transistor as an amplifier.
- Develop a strong understanding of common base (CB) transistor connection, common emitter (CE) transistor connection and common collector (CC) transistor connection.
- Understand the input and output characteristics of common base (CB) transistor connection and common emitter (CE) transistor connection.
- Explore the reasons for the widespread use of common emitter (CE) transistor connection.
- Develop an understanding of transistor load line analysis.
- Understand the use of transistor as constant current source.
- Learn the importance of power rating of transistor.
- Explore the applications of transistors.

Introduction

When a third doped element is added to a semiconductor diode in such a way that two *pn* junctions are formed, the resulting device is known as a *bipolar junction transistor (BJT)* or simply *transistor*. Thus a semiconductor diode has one *pn* junction while a transistor has two *pn* junctions. The primary function of a transistor is to raise the strength of a weak signal *i.e.* it acts as an amplifier. Transistors find many other applications in electronic circuits. For example, a transistor is used as a high-speed electronic switch in digital circuits. In this chapter, we shall focus our attention on various aspects of transistors and their increasing applications in electronics industry.

6.1 Transistor

A **transistor** consists of two *pn* junctions formed by **sandwiching either p-type or n-type semiconductor between a pair of opposite types*. Accordingly; there are two types of transistors, namely ;

(i) *n-p-n* transistor

(ii) *p-n-p* transistor

* In practice, these three blocks *p, n, p* are grown out of the same crystal by adding corresponding impurities in turn.

An n - p - n transistor is composed of two n -type semiconductors separated by a thin section of p -type as shown in Fig. 6.1 (i). However, a p - n - p transistor is formed by two p -sections separated by a thin section of n -type as shown in Fig. 6.1 (ii).

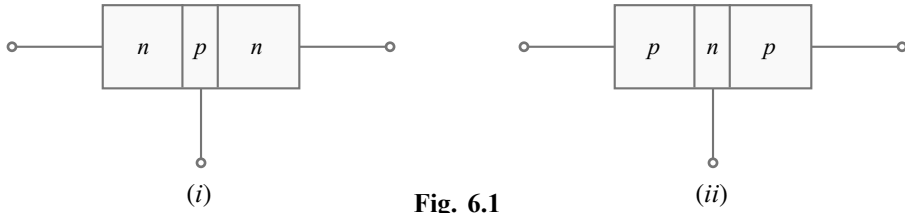


Fig. 6.1

In each type of transistor, the following points may be noted :

- (i) There are two pn junctions. Therefore, a transistor may be regarded as a combination of two diodes connected back to back.
- (ii) There are three terminals, one taken from each type of semiconductor.
- (iii) The middle section is a very thin layer. This is the most important factor in the function of a transistor.

Origin of the name “Transistor”. When new devices are invented, scientists often try to devise a name that will appropriately describe the device. A transistor has two pn junctions. As discussed later, one junction is forward biased and the other is reverse biased. The forward biased junction has a low resistance path whereas a reverse biased junction has a high resistance path. The weak signal is introduced in the low resistance circuit and output is taken from the high resistance circuit. Therefore, a transistor *transfers* a signal from a low resistance to high resistance. The prefix ‘trans’ means the signal transfer property of the device while ‘istor’ classifies it as a solid element in the same general family with resistors.

Note. The two junction transistor shown in Fig. 6.1 is called bipolar junction transistor (*BJT*) because two (Bi means two) polarities charge carriers (*viz.* positively charged holes and negatively charged free electrons) are involved in its operation. *Note that in this chapter when we use the terms transistor, it means bipolar junction transistor (BJT) unless stated otherwise.*

6.2 Naming the Transistor Terminals

A transistor (pnp or npn) has three sections of doped semiconductors. The section on one side is the *emitter* and the section on the opposite side is the *collector*. The middle section is called the *base* and forms two junctions between the emitter and collector.

(i) **Emitter.** The section on one side that supplies charge carriers (electrons or holes) is called the *emitter*. *The emitter is always forward biased w.r.t. base* so that it can supply a large number of *majority carriers. In Fig. 6.2 (i), the emitter (p -type) of pnp transistor is forward biased and supplies hole charges to its junction with the base. Similarly, in Fig. 6.2 (ii), the emitter (n -type) of npn transistor has a forward bias and supplies free electrons to its junction with the base.

(ii) **Collector.** The section on the other side that collects the charges is called the *collector*. *The collector is always reverse biased.* Its function is to remove charges from its junction with the base. In Fig. 6.2 (i), the collector (p -type) of pnp transistor has a reverse bias and receives hole charges that flow in the output circuit. Similarly, in Fig. 6.2 (ii), the collector (n -type) of npn transistor has reverse bias and receives electrons.

* Holes if emitter is p -type and electrons if the emitter is n -type.

$$\text{Input current, } I_E = \frac{\text{Signal}}{R_{in}} = \frac{500 \text{ mV}}{20 \Omega} = 25 \text{ mA}$$

Since α_{ac} is nearly 1, output current, $I_C = I_E = 25 \text{ mA}$.

$$\text{Output voltage, } V_{out} = I_C R_C = 25 \text{ mA} \times 1 \text{ k}\Omega = 25 \text{ V}$$

$$\therefore \text{Voltage amplification, } A_v = \frac{V_{out}}{\text{Signal}} = \frac{25 \text{ V}}{500 \text{ mV}} = 50$$

Comments. The reader may note that basic amplifying action is produced by transferring a current from a *low-resistance* to a *high-resistance* circuit. Consequently, the name transistor is given to the device by combining the two terms given in magenta letters below :

Transfer + Resistor $\longrightarrow$ Transistor

6.8 Transistor Connections

There are three leads in a transistor viz., emitter, base and collector terminals. However, when a transistor is to be connected in a circuit, we require four terminals ; two for the input and two for the output. This difficulty is overcome by making one terminal of the transistor common to both input and output terminals. The input is fed between this common terminal and one of the other two terminals. The output is obtained between the common terminal and the remaining terminal. Accordingly ; a transistor can be connected in a circuit in the following three ways :

- (i) Common base connection
- (ii) Common emitter connection
- (iii) Common collector connection

Each circuit connection has specific advantages and disadvantages. It may be noted here that regardless of circuit connection, the emitter is always biased in the forward direction, while the collector always has a reverse bias.

6.9 Common Base Connection

In this circuit arrangement, input is applied between emitter and base and output is taken from collector and base. Here, base of the transistor is common to both input and output circuits and hence the name common base connection. In Fig. 6.10 (i), a common base *npn* transistor circuit is shown whereas Fig. 6.10 (ii) shows the common base *pnp* transistor circuit.

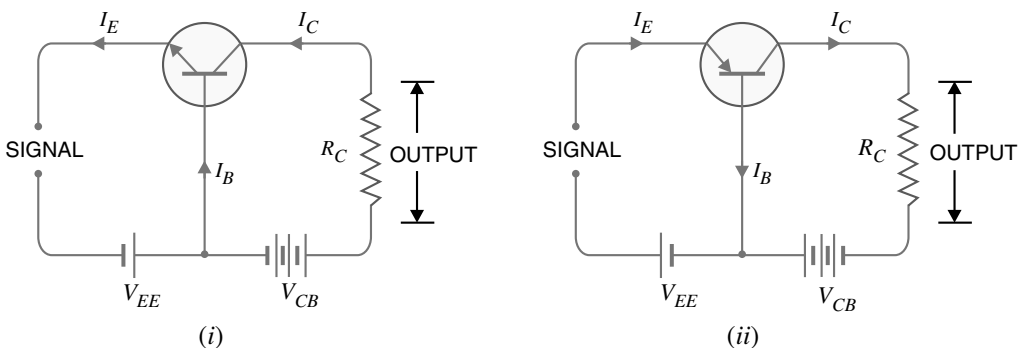


Fig. 6.10

1. Current amplification factor (α). It is the ratio of output current to input current. In a common base connection, the input current is the emitter current I_E and output current is the collector current I_C .

The ratio of change in collector current to the change in emitter current at constant collector-base voltage V_{CB} is known as **current amplification factor** i.e.

$$*\alpha = \frac{\Delta I_C}{\Delta I_E} \text{ at constant } V_{CB}$$

It is clear that current amplification factor is less than **unity. This value can be increased (but not more than unity) by decreasing the base current. This is achieved by making the base thin and doping it lightly. Practical values of α in commercial transistors range from 0.9 to 0.99.

2. Expression for collector current. The whole of emitter current does not reach the collector. It is because a small percentage of it, as a result of electron-hole combinations occurring in base area, gives rise to base current. Moreover, as the collector-base junction is reverse biased, therefore, some leakage current flows due to minority carriers. It follows, therefore, that total collector current consists of :

(i) That part of emitter current which reaches the collector terminal i.e. *** αI_E .

(ii) The leakage current $I_{leakage}$. This current is due to the movement of minority carriers across base-collector junction on account of it being reverse biased. This is generally much smaller than αI_E .

$$\therefore \text{Total collector current, } I_C = \alpha I_E + I_{leakage}$$

It is clear that if $I_E = 0$ (i.e. emitter circuit is open), a small leakage current still flows in the collector circuit. This $I_{leakage}$ is abbreviated as I_{CBO} , meaning collector-base current with emitter open. The I_{CBO} is indicated in Fig. 6.11.

$$\therefore I_C = \alpha I_E + I_{CBO} \quad \dots(i)$$

$$\text{Now, } I_E = I_C + I_B$$

$$\therefore I_C = \alpha (I_C + I_B) + I_{CBO}$$

$$\text{or } I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\text{or } I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha} \quad \dots(ii)$$

Relation (i) or (ii) can be used to find I_C . It is further clear from these relations that the collector current of a transistor can be controlled by either the emitter or base current.

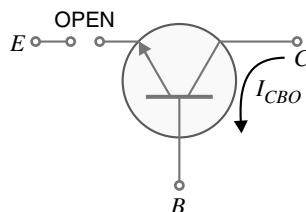


Fig. 6.11

* If only d.c. values are considered, then, $\alpha = I_C/I_E$.

** At first sight, it might seem that since there is no current gain, no voltage or power amplification could be possible with this arrangement. However, it may be recalled that output circuit resistance is much higher than the input circuit resistance. Therefore, it does give rise to voltage and power gain.

*** $\alpha = \frac{I_C}{I_E} \quad \therefore I_C = \alpha I_E$

In other words, αI_E part of emitter current reaches the collector terminal.

Fig. 6.12 shows the concept of I_{CBO} . In CB configuration, a small collector current flows even when the emitter current is zero. This is the leakage collector current (*i.e.* the collector current when emitter is open) and is denoted by I_{CBO} . When the emitter voltage V_{EE} is also applied, the various currents are as shown in Fig. 6.12 (ii).

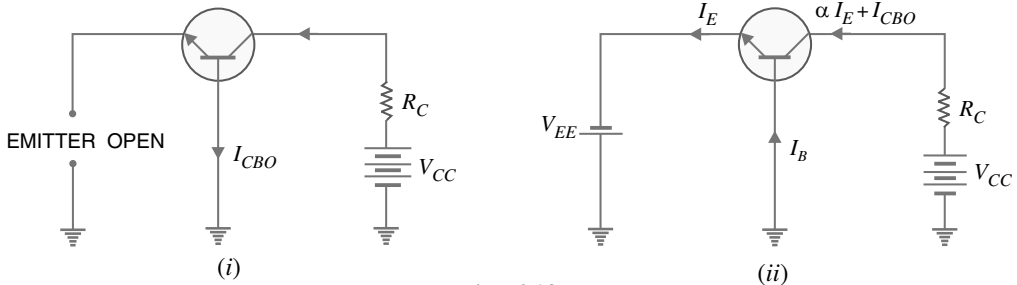


Fig. 6.12

Note. Owing to improved construction techniques, the magnitude of I_{CBO} for general-purpose and low-powered transistors (especially silicon transistors) is usually very small and may be neglected in calculations. However, for high power applications, it will appear in microampere range. Further, I_{CBO} is very much temperature dependent ; it increases rapidly with the increase in temperature. Therefore, at higher temperatures, I_{CBO} plays an important role and must be taken care of in calculations.

Example 6.6. In a common base connection, current amplification factor is 0.9. If the emitter current is 1mA, determine the value of base current.

Solution. Here, $\alpha = 0.9$; $I_E = 1 \text{ mA}$

$$\text{Now,} \quad \alpha = \frac{I_C}{I_E}$$

$$\text{or} \quad I_C = \alpha I_E = 0.9 \times 1 = 0.9 \text{ mA}$$

$$\text{Also,} \quad I_E = I_B + I_C$$

$$\therefore \quad \text{Base current, } I_B = I_E - I_C = 1 - 0.9 = 0.1 \text{ mA}$$

Example 6.7. In common base connection, $\alpha = 0.98$ and the base current is $100 \mu\text{A}$. Find the collector current and emitter current.

$$\text{Solution.} \quad I_E = I_B + I_C$$

$$\text{Now,} \quad I_C = \alpha I_E = \alpha(I_B + I_C)$$

$$\text{or} \quad I_C(1 - \alpha) = \alpha I_B$$

$$\therefore \quad I_C = \frac{\alpha I_B}{1 - \alpha} = \frac{0.98 \times 100 \mu\text{A}}{1 - 0.98} = 4900 \mu\text{A} = 4.9 \text{ mA}$$

$$\text{Now,} \quad I_E = \frac{I_C}{\alpha} = \frac{4.9 \text{ mA}}{0.98} = 5 \text{ mA}$$

$$\text{Alternatively,} \quad I_E = I_B + I_C = 100 \mu\text{A} + 4.9 \text{ mA} = 5 \text{ mA}$$

Example 6.8. The emitter current in a common base npn transistor circuit is 8.4 mA. If 0.8% of free electrons injected into the base recombine with holes and the leakage current is $0.1 \mu\text{A}$, find (i) base current (ii) collector current (iii) exact value of α (iv) approximate value of α , neglecting I_{CBO} .

Solution. It is given : $I_E = 8.4 \text{ mA}$; $I_{CBO} = 0.1 \mu\text{A} = 10^{-7} \text{ mA}$

(i) $I_B = 0.8\% \text{ of } I_E = 0.008 \times 8.4 \text{ mA} = 0.0672 \text{ mA}$

(ii) $I_C = I_E - I_B = 8.4 - 0.0672 = 8.3328 \text{ mA}$

(iii) Now, $\alpha_{I_E} = I_C - I_{CBO} = 8.3328 - 10^{-7} = 8.3327 \text{ mA}$

$\therefore$ Exact value of $\alpha = \frac{8.3327}{8.4} = 0.9919$

(iv) Approximate $\alpha = \frac{I_C}{I_E} = \frac{8.3328}{8.4} = 0.992$

Note that the difference between the exact and approximate values of α is so small that it can be neglected without any appreciable error.

Example 6.9. In a common base connection, the emitter current is 1 mA. If the emitter circuit is open, the collector current is 50 μA . Find the total collector current. Given that $\alpha = 0.92$.

Solution. Here, $I_E = 1 \text{ mA}$; $\alpha = 0.92$; $I_{CBO} = 50 \mu\text{A}$

$\therefore$ Total collector current, $I_C = \alpha I_E + I_{CBO} = 0.92 \times 1 + 50 \times 10^{-3}$
 $= 0.92 + 0.05 = 0.97 \text{ mA}$

Example 6.10. In a common base connection, $\alpha = 0.95$. The voltage drop across 2 k Ω resistance which is connected in the collector is 2 V. Find the base current.

Solution. Fig. 6.13 shows the required common base connection. The voltage drop across $R_C (= 2 \text{ k}\Omega)$ is 2 V.

$\therefore I_C = 2\text{V}/2\text{k}\Omega = 1 \text{ mA}$

Now, $\alpha = I_C/I_E$

$\therefore I_E = \frac{I_C}{\alpha} = \frac{1}{0.95} = 1.05 \text{ mA}$

Using the relation, $I_E = I_B + I_C$

$\therefore I_B = I_E - I_C = 1.05 - 1$
 $= 0.05 \text{ mA}$

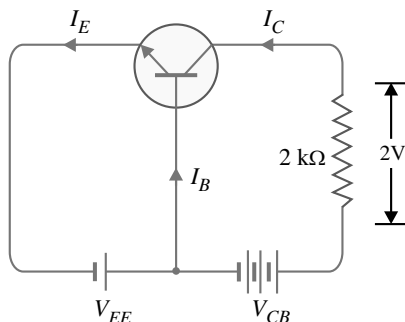


Fig. 6.13

Example 6.11. For the common base circuit shown in Fig. 6.14, determine I_C and V_{CB} . Assume the transistor to be of silicon.

Solution. Since the transistor is of silicon, $V_{BE} = 0.7 \text{ V}$. Applying Kirchhoff's voltage law to the emitter-side loop, we get,

$$V_{EE} = I_E R_E + V_{BE}$$

or $I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{8\text{V} - 0.7\text{V}}{1.5 \text{ k}\Omega} = 4.87 \text{ mA}$

$\therefore I_C \simeq I_E = 4.87 \text{ mA}$

Applying Kirchhoff's voltage law to the collector-side loop, we have,

$$V_{CC} = I_C R_C + V_{CB}$$

$$\begin{aligned} \therefore V_{CB} &= V_{CC} - I_C R_C \\ &= 18 \text{ V} - 4.87 \text{ mA} \times 1.2 \text{ k}\Omega = 12.16 \text{ V} \end{aligned}$$

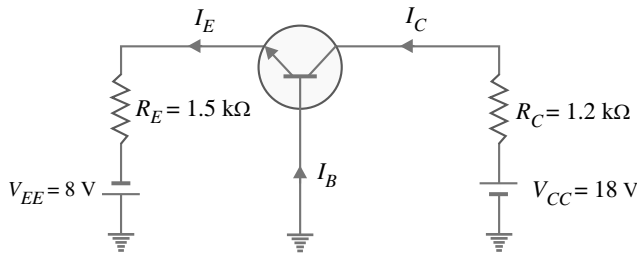


Fig. 6.14

6.10 Characteristics of Common Base Connection

The complete electrical behaviour of a transistor can be described by stating the interrelation of the various currents and voltages. These relationships can be conveniently displayed graphically and the curves thus obtained are known as the characteristics of transistor. The most important characteristics of common base connection are *input characteristics* and *output characteristics*.

1. Input characteristic. It is the curve between emitter current I_E and emitter-base voltage V_{EB} at constant collector-base voltage V_{CB} . The emitter current is generally taken along y -axis and emitter-base voltage along x -axis. Fig. 6.15 shows the input characteristics of a typical transistor in CB arrangement. The following points may be noted from these characteristics:

(i) The emitter current I_E increases rapidly with small increase in emitter-base voltage V_{EB} . It means that input resistance is very small.

(ii) The emitter current is almost independent of collector-base voltage V_{CB} . This leads to the conclusion that emitter current (and hence collector current) is almost independent of collector voltage.

Input resistance. It is the ratio of change in emitter-base voltage (ΔV_{EB}) to the resulting change in emitter current (ΔI_E) at constant collector-base voltage (V_{CB}) i.e.

$$\text{Input resistance, } r_i = \frac{\Delta V_{BE}}{\Delta I_E} \text{ at constant } V_{CB}$$

In fact, input resistance is the opposition offered to the signal current. As a very small V_{EB} is sufficient to produce a large flow of emitter current I_E , therefore, input resistance is quite small, of the order of a few ohms.

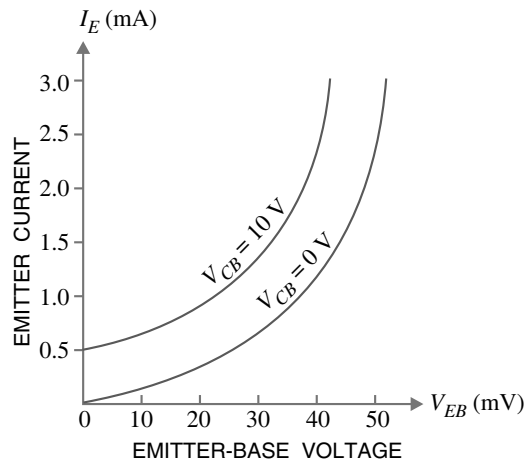


Fig. 6.15

2. Output characteristic. It is the curve between collector current I_C and collector-base voltage V_{CB} at *constant emitter current I_E . Generally, collector current is taken along y-axis and collector-base voltage along x-axis. Fig. 6.16 shows the output characteristics of a typical transistor in CB arrangement.

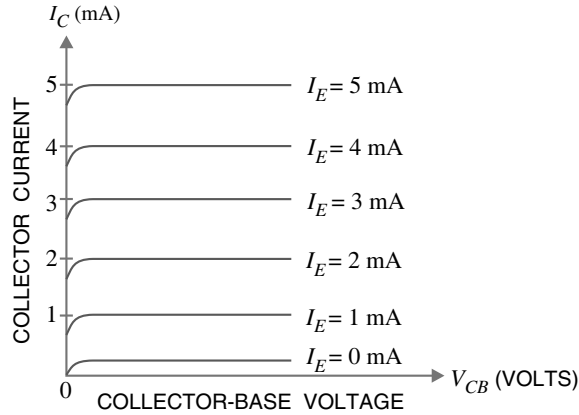


Fig. 6.16

The following points may be noted from the characteristics :

(i) The collector current I_C varies with V_{CB} only at very low voltages (< 1 V). The transistor is *never* operated in this region.

(ii) When the value of V_{CB} is raised above 1 – 2 V, the collector current becomes constant as indicated by straight horizontal curves. It means that now I_C is independent of V_{CB} and depends upon I_E only. This is consistent with the theory that the emitter current flows *almost* entirely to the collector terminal. The transistor is *always* operated in this region.

(iii) A very large change in collector-base voltage produces only a tiny change in collector current. This means that output resistance is very high.

Output resistance. It is the ratio of change in collector-base voltage (ΔV_{CB}) to the resulting change in collector current (ΔI_C) at constant emitter current *i.e.*

$$\text{Output resistance, } r_o = \frac{\Delta V_{CB}}{\Delta I_C} \text{ at constant } I_E$$

The output resistance of CB circuit is very high, of the order of several tens of kilo-ohms. This is not surprising because the collector current changes very slightly with the change in V_{CB} .

6.11 Common Emitter Connection

In this circuit arrangement, input is applied between base and emitter and output is taken from the collector and emitter. Here, emitter of the transistor is common to both input and output

* I_E has to be kept constant because any change in I_E will produce corresponding change in I_C . Here, we are interested to see how V_{CB} influences I_C .

circuits and hence the name common emitter connection. Fig. 6.17 (i) shows common emitter *nnp* transistor circuit whereas Fig. 6.17 (ii) shows common emitter *pnp* transistor circuit.

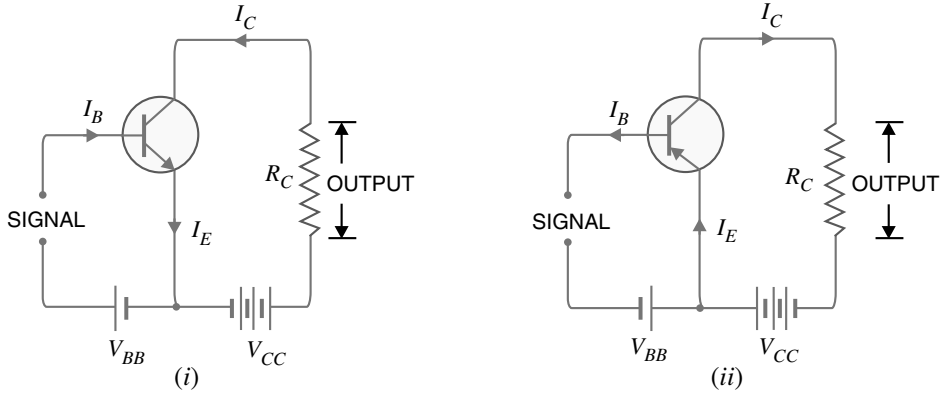


Fig. 6.17

1. Base current amplification factor (β). In common emitter connection, input current is I_B and output current is I_C .

The ratio of change in collector current (ΔI_C) to the change in base current (ΔI_B) is known as **base current amplification factor** i.e.

$$\beta^* = \frac{\Delta I_C}{\Delta I_B}$$

In almost any transistor, less than 5% of emitter current flows as the base current. Therefore, the value of β is generally greater than 20. Usually, its value ranges from 20 to 500. This type of connection is frequently used as it gives appreciable current gain as well as voltage gain.

Relation between β and α . A simple relation exists between β and α . This can be derived as follows :

$$\beta = \frac{\Delta I_C}{\Delta I_B} \quad \dots(i)$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \quad \dots(ii)$$

Now,

$$I_E = I_B + I_C$$

or

$$\Delta I_E = \Delta I_B + \Delta I_C$$

or

$$\Delta I_B = \Delta I_E - \Delta I_C$$

Substituting the value of ΔI_B in exp. (i), we get,

$$\beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C} \quad \dots(iii)$$

Dividing the numerator and denominator of R.H.S. of exp. (iii) by ΔI_E , we get,

$$\beta = \frac{\frac{\Delta I_C}{\Delta I_E}}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{\alpha}{1 - \alpha} \quad \left[\because \alpha = \frac{\Delta I_C}{\Delta I_E} \right]$$

* If d.c. values are considered, $\beta = I_C/I_B$.

$$\therefore \beta = \frac{\alpha}{1 - \alpha}$$

It is clear that as α approaches unity, β approaches infinity. In other words, the current gain in common emitter connection is very high. It is due to this reason that this circuit arrangement is used in about 90 to 95 percent of all transistor applications.

2. Expression for collector current. In common emitter circuit, I_B is the input current and I_C is the output current.

$$\text{We know } I_E = I_B + I_C \quad \dots(i)$$

$$\text{and } I_C = \alpha I_E + I_{CBO} \quad \dots(ii)$$

$$\text{From exp. (ii), we get, } I_C = \alpha I_E + I_{CBO} = \alpha (I_B + I_C) + I_{CBO}$$

$$\text{or } I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\text{or } I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO} \quad \dots(iii)$$

From exp. (iii), it is apparent that if $I_B = 0$ (i.e. base circuit is open), the collector current will be the current to the emitter. This is abbreviated as I_{CEO} , meaning collector-emitter current with base open.

$$\therefore I_{CEO} = \frac{1}{1 - \alpha} I_{CBO}$$

Substituting the value of $\frac{1}{1 - \alpha} I_{CBO} = I_{CEO}$ in exp. (iii), we get,

$$I_C = \frac{\alpha}{1 - \alpha} I_B + I_{CEO}$$

$$\text{or } I_C = \beta I_B + I_{CEO} \quad \left(\because \beta = \frac{\alpha}{1 - \alpha} \right)$$

Concept of I_{CEO} . In CE configuration, a small collector current flows even when the base current is zero [See Fig. 6.18 (i)]. This is the collector cut-off current (i.e. the collector current that flows when base is open) and is denoted by I_{CEO} . The value of I_{CEO} is much larger than I_{CBO} .

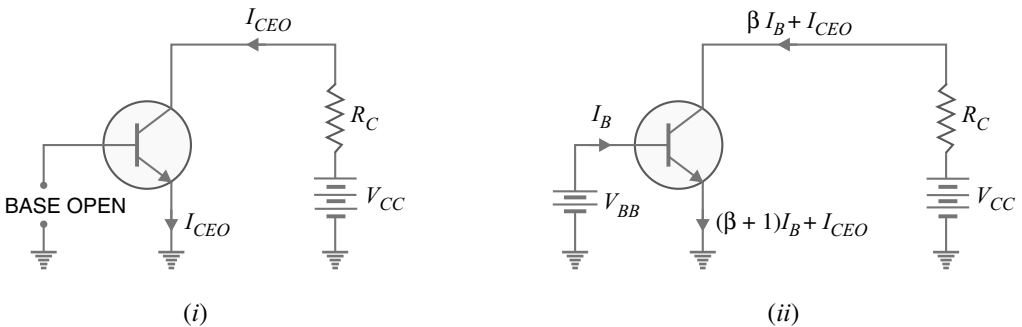


Fig. 6.18

When the base voltage is applied as shown in Fig. 6.18 (ii), then the various currents are :

$$\text{Base current} = I_B$$

$$\text{Collector current} = \beta I_B + I_{CEO}$$

$$\begin{aligned}\text{Emitter current} &= \text{Collector current} + \text{Base current} \\ &= (\beta I_B + I_{CEO}) + I_B = (\beta + 1) I_B + I_{CEO}\end{aligned}$$

$$\text{It may be noted here that : } I_{CEO} = \frac{1}{1-\alpha} I_{CBO} = (\beta + 1) I_{CBO} \left[\because \frac{1}{1-\alpha} = \beta + 1 \right]$$

6.12 Measurement of Leakage Current

A very small leakage current flows in all transistor circuits. However, in most cases, it is quite small and can be neglected.

(i) **Circuit for I_{CEO} test.** Fig. 6.19 shows the circuit for measuring I_{CEO} . Since base is open ($I_B = 0$), the transistor is in cut-off. Ideally, $I_C = 0$ but actually there is a small current from collector to emitter due to minority carriers. It is called I_{CEO} (collector-to-emitter current with base open). This current is usually in the nA range for silicon. A faulty transistor will often have excessive leakage current.

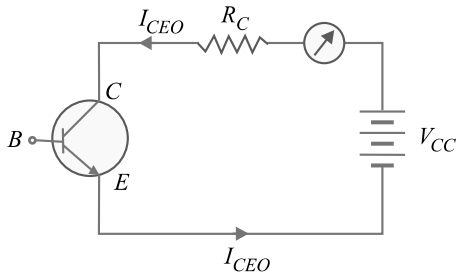


Fig. 6.19

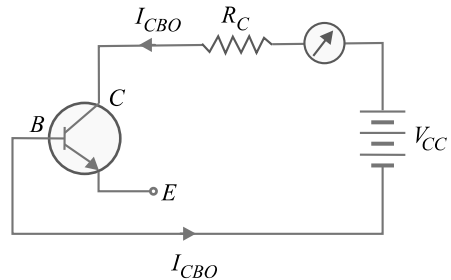


Fig. 6.20

(ii) **Circuit for I_{CBO} test.** Fig. 6.20 shows the circuit for measuring I_{CBO} . Since the emitter is open ($I_E = 0$), there is a small current from collector to base. This is called I_{CBO} (collector-to-base current with emitter open). This current is due to the movement of minority carriers across base-collector junction. The value of I_{CBO} is also small. If in measurement, I_{CBO} is excessive, then there is a possibility that collector-base is shorted.

Example 6.12. A transistor has $I_{CBO} = 48 \text{ nA}$ and $\alpha = 0.992$. Find (i) β and I_{CEO} (ii) collector current when $I_B = 30 \mu\text{A}$.

Solution.

$$(i) \quad \beta = \frac{\alpha}{1-\alpha} = \frac{0.992}{1-0.992} = 124$$

$$I_{CEO} = \frac{I_{CBO}}{1-\alpha} = \frac{48 \times 10^{-9} \text{ A}}{1-0.992} = 6 \times 10^{-6} \text{ A} = 6 \mu\text{A}$$

Alternatively,

$$I_{CEO} = (\beta + 1) I_{CBO} = (124 + 1) \times 48 \times 10^{-3} \mu\text{A} = 6 \mu\text{A}$$

$$(ii) \quad I_C = \beta I_B + I_{CEO} = 124 \times 30 \mu\text{A} + 6 \mu\text{A} = 3726 \mu\text{A} = 3.726 \text{ mA}$$

Example 6.13. A transistor has $I_{CBO} = 0.1 \mu\text{A}$ and $I_{CEO} = 16 \mu\text{A}$. Find its α .

$$\begin{aligned}\text{Solution.} \quad I_{CEO} &= \frac{I_{CBO}}{1-\alpha} \\ \text{or} \quad 1-\alpha &= \frac{I_{CBO}}{I_{CEO}} = \frac{0.1 \mu\text{A}}{16 \mu\text{A}} = 0.00625 \\ \therefore \quad \alpha &= 1 - 0.00625 = 0.99375\end{aligned}$$

Example 6.14. Find the α rating of the transistor shown in Fig. 6.21. Hence determine the value of I_C using both α and β ratings of the transistor.

Solution. Fig. 6.21 shows the conditions of the problem.

$$\alpha = \frac{\beta}{1 + \beta} = \frac{49}{1 + 49} = 0.98$$

The value of I_C can be found by using either α or β rating as under :

$$I_C = \alpha I_E = 0.98 (12 \text{ mA}) = 11.76 \text{ mA}$$

$$\text{Also, } I_C = \beta I_B = 49 (240 \mu\text{A}) = 11.76 \text{ mA}$$

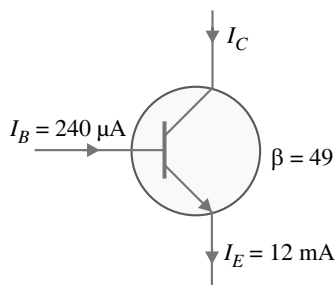


Fig. 6.21

Example 6.15. A transistor is connected in common emitter (CE) configuration in which collector supply is 8V and the voltage drop across resistance R_C connected in the collector circuit is 0.5 V. The value of $R_C = 800 \Omega$. If $\alpha = 0.96$, determine :

- collector-emitter voltage
- base current

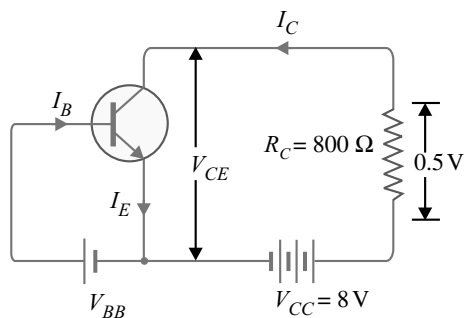


Fig. 6.22

Solution. Fig. 6.22 shows the required common emitter connection with various values.

- Collector-emitter voltage is

$$V_{CE} = V_{CC} - 0.5 = 8 - 0.5 = 7.5 \text{ V}$$

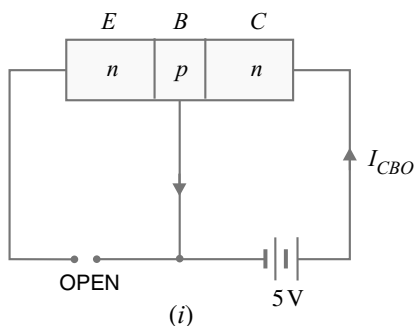
- The voltage drop across $R_C (= 800 \Omega)$ is 0.5 V.

$$\therefore I_C = \frac{0.5 \text{ V}}{800 \Omega} = \frac{5}{8} \text{ mA} = 0.625 \text{ mA}$$

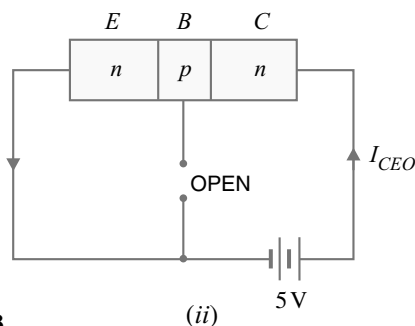
$$\text{Now, } \beta = \frac{\alpha}{1 - \alpha} = \frac{0.96}{1 - 0.96} = 24$$

$$\therefore \text{Base current, } I_B = \frac{I_C}{\beta} = \frac{0.625}{24} = 0.026 \text{ mA}$$

Example 6.16. An n-p-n transistor at room temperature has its emitter disconnected. A voltage of 5 V is applied between collector and base. With collector positive, a current of $0.2 \mu\text{A}$ flows. When the base is disconnected and the same voltage is applied between collector and emitter, the current is found to be $20 \mu\text{A}$. Find α , I_E and I_B when collector current is 1mA.



(i)



(ii)

Fig. 6.23

Solution. When the emitter circuit is open [See Fig. 6.23 (i)], the collector-base junction is reverse biased. A small leakage current I_{CBO} flows due to minority carriers.

$$\therefore I_{CBO} = 0.2 \mu\text{A} \quad \dots \text{given}$$

When base is open [See Fig. 6.23 (ii)], a small leakage current I_{CEO} flows due to minority carriers.

$$\therefore I_{CEO} = 20 \mu\text{A} \quad \dots \text{given}$$

$$\text{We know : } I_{CEO} = \frac{I_{CBO}}{1 - \alpha}$$

$$\text{or } 20 = \frac{0.2}{1 - \alpha}$$

$$\therefore \alpha = 0.99$$

$$\text{Now, } I_C = \alpha I_E + I_{CBO}$$

$$\text{Here, } I_C = 1 \text{ mA} = 1000 \mu\text{A} ; \alpha = 0.99 ; I_{CBO} = 0.2 \mu\text{A}$$

$$\therefore 1000 = 0.99 \times I_E + 0.2$$

$$\text{or } I_E = \frac{1000 - 0.2}{0.99} = 1010 \mu\text{A}$$

$$\text{and } I_B = I_E - I_C = 1010 - 1000 = 10 \mu\text{A}$$

Example 6.17. The collector leakage current in a transistor is $300 \mu\text{A}$ in CE arrangement. If now the transistor is connected in CB arrangement, what will be the leakage current? Given that $\beta = 120$.

$$\begin{aligned} \text{Solution. } I_{CEO} &= 300 \mu\text{A} \\ \beta &= 120 ; \alpha = \frac{\beta}{\beta + 1} = \frac{120}{120 + 1} = 0.992 \end{aligned}$$

$$\text{Now, } I_{CEO} = \frac{I_{CBO}}{1 - \alpha}$$

$$\therefore I_{CBO} = (1 - \alpha) I_{CEO} = (1 - 0.992) \times 300 = 2.4 \mu\text{A}$$

Note that leakage current in CE arrangement (i.e. I_{CEO}) is much more than in CB arrangement (i.e. I_{CBO}).

Example 6.18. For a certain transistor, $I_B = 20 \mu\text{A}$; $I_C = 2 \text{ mA}$ and $\beta = 80$. Calculate I_{CBO} .

Solution.

$$I_C = \beta I_B + I_{CEO}$$

$$\text{or } 2 = 80 \times 0.02 + I_{CEO}$$

$$\therefore I_{CEO} = 2 - 80 \times 0.02 = 0.4 \text{ mA}$$

$$\text{Now, } \alpha = \frac{\beta}{\beta + 1} = \frac{80}{80 + 1} = 0.988$$

$$\therefore I_{CBO} = (1 - \alpha) I_{CEO} = (1 - 0.988) \times 0.4 = 0.0048 \text{ mA}$$

Example 6.19. Using diagrams, explain the correctness of the relation $I_{CEO} = (\beta + 1) I_{CBO}$

Solution. The leakage current I_{CBO} is the current that flows through the base-collector junction when emitter is open as shown in Fig. 6.24. When the transistor is in *CE* arrangement, the *base current (i.e. I_{CBO}) is multiplied by β in the collector as shown in Fig. 6.25.

$$\therefore I_{CEO} = I_{CBO} + \beta I_{CBO} = (\beta + 1) I_{CBO}$$

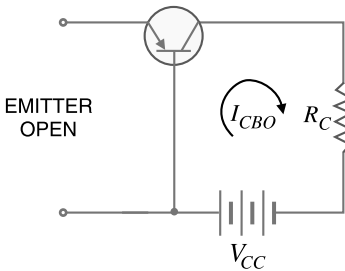


Fig. 6.24

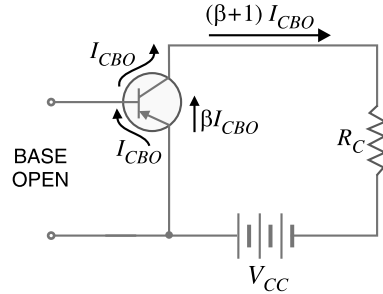


Fig. 6.25

Example 6.20. Determine V_{CB} in the transistor **circuit shown in Fig. 6.26 (i). The transistor is of silicon and has $\beta = 150$.

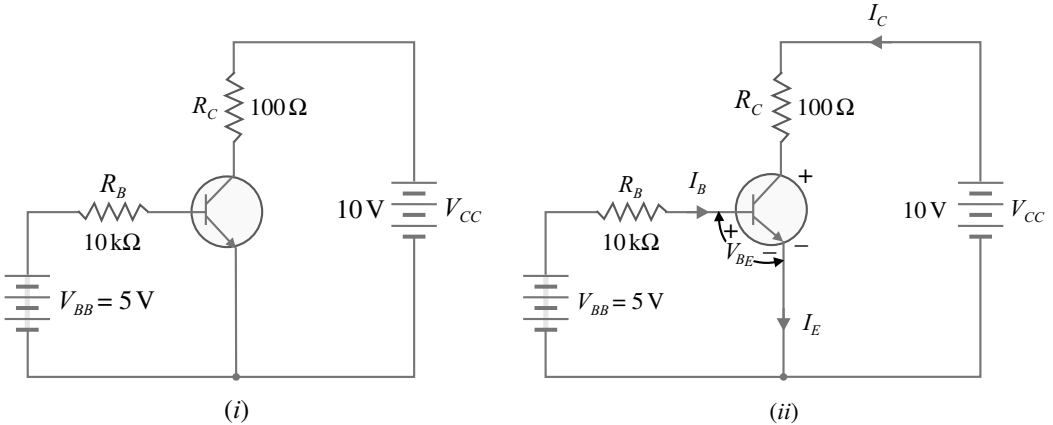


Fig. 6.26

Solution. Fig. 6.26 (i) shows the transistor circuit while Fig. 6.26 (ii) shows the various currents and voltages alongwith polarities.

Applying Kirchhoff's voltage law to base-emitter loop, we have,

$$V_{BB} - I_B R_B - V_{BE} = 0$$

$$\text{or } I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5\text{V} - 0.7\text{V}}{10\text{k}\Omega} = 430\text{ }\mu\text{A}$$

$$\therefore I_C = \beta I_B = (150) (430\text{ }\mu\text{A}) = 64.5\text{ mA}$$

* The current I_{CBO} is amplified because it is forced to flow across the base-emitter junction.

** The resistor R_B controls the base current I_B and hence collector current $I_C (= \beta I_B)$. If R_B is increased, the base current (I_B) decreases and hence collector current (I_C) will decrease and vice-versa.

$$\begin{aligned}\text{Now, } V_{CE} &= V_{CC} - I_C R_C \\ &= 10 \text{ V} - (64.5 \text{ mA}) (100 \Omega) = 10 \text{ V} - 6.45 \text{ V} = 3.55 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{We know that : } V_{CE} &= V_{CB} + V_{BE} \\ \therefore V_{CB} &= V_{CE} - V_{BE} = 3.55 - 0.7 = 2.85 \text{ V}\end{aligned}$$

Example 6.21. In a transistor, $I_B = 68 \mu\text{A}$, $I_E = 30 \text{ mA}$ and $\beta = 440$. Determine the α rating of the transistor. Then determine the value of I_C using both the α rating and β rating of the transistor.

Solution.

$$\alpha = \frac{\beta}{\beta + 1} = \frac{440}{440 + 1} = 0.9977$$

$$I_C = \alpha I_E = (0.9977) (30 \text{ mA}) = 29.93 \text{ mA}$$

$$\text{Also, } I_C = \beta I_B = (440) (68 \mu\text{A}) = 29.93 \text{ mA}$$

Example 6.22. A transistor has the following ratings : $I_{C(\text{max})} = 500 \text{ mA}$ and $\beta_{(\text{max})} = 300$. Determine the maximum allowable value of I_B for the device.

Solution.

$$I_{B(\text{max})} = \frac{I_{C(\text{max})}}{\beta_{(\text{max})}} = \frac{500 \text{ mA}}{300} = 1.67 \text{ mA}$$

For this transistor, if the base current is allowed to exceed 1.67 mA, the collector current will exceed its maximum rating of 500 mA and the transistor will probably be destroyed.

Example 6.23. Fig. 6.27 shows the open-circuit failures in a transistor. What will be the circuit behaviour in each case?

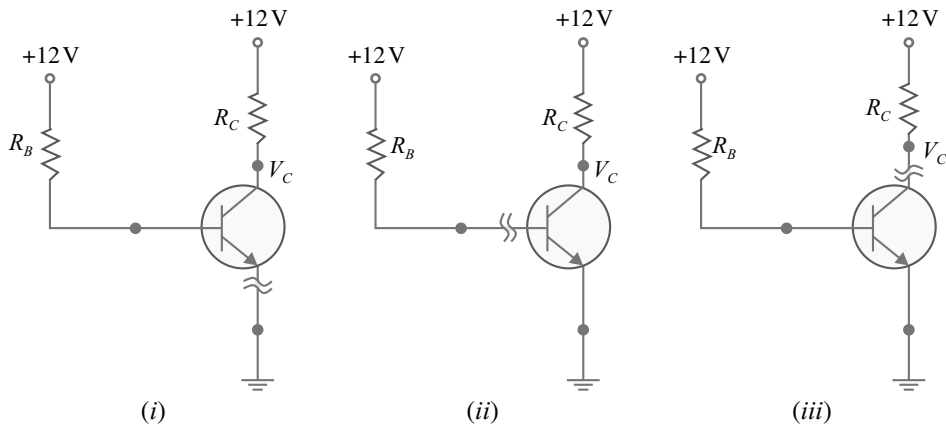


Fig. 6.27

Solution. *Fig. 6.27 shows the open-circuit failures in a transistor. We shall discuss the circuit behaviour in each case.

(i) Open emitter. Fig. 6.27 (i) shows an open emitter failure in a transistor. Since the collector diode is not forward biased, it is *OFF* and there can be neither collector current nor base current. Therefore, there will be no voltage drops across either resistor and the voltage at the base and at the collector leads of the transistor will be 12 V.

* The collector resistor R_C controls the collector voltage $V_C (= V_{CC} - I_C R_C)$. When R_C increases, V_C decreases and *vice-versa*.

(ii) **Open base.** Fig. 6.27 (ii) shows an open base failure in a transistor. Since the base is open, there can be no base current so that the transistor is in *cut-off*. Therefore, all the transistor currents are 0A. In this case, the base and collector voltages will both be at 12 V.

Note. It may be noted that an open failure at either the base or emitter will produce similar results.

(iii) **Open collector.** Fig. 6.27 (iii) shows an open collector failure in a transistor. In this case, the emitter diode is still *ON*, so we expect to see 0.7 V at the base. However, we will see 12 V at the collector because there is no collector current.

Example 6.24. Fig. 6.28 shows the short-circuit failures in a transistor. What will be the circuit behaviour in each case?

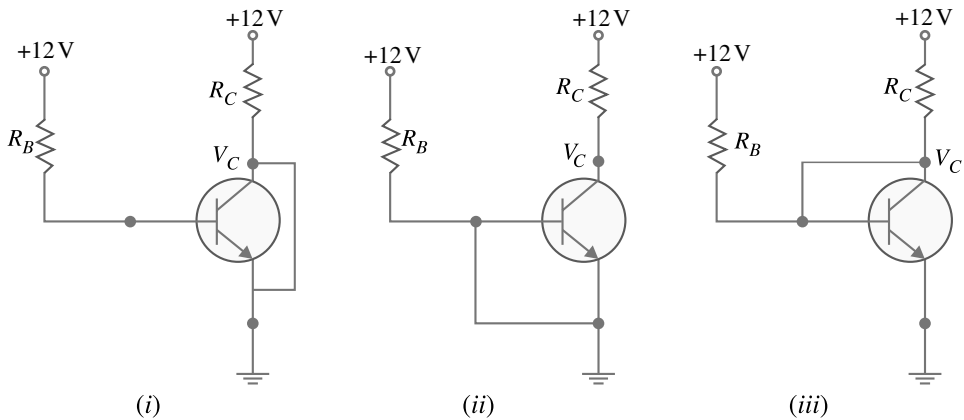


Fig. 6.28

Solution. Fig. 6.28 shows the short-circuit failures in a transistor. We shall discuss the circuit behaviour in each case.

(i) **Collector-emitter short.** Fig. 6.28 (i) shows a short between collector and emitter. The emitter diode is still forward biased, so we expect to see 0.7 V at the base. Since the collector is shorted to the emitter, $V_C = V_E = 0$ V.

(ii) **Base-emitter short.** Fig. 6.28 (ii) shows a short between base and emitter. Since the base is now directly connected to ground, $V_B = 0$. Therefore, the current through R_B will be diverted to ground and there is no current to forward bias the emitter diode. As a result, the transistor will be *cut-off* and there is no collector current. So we will expect the collector voltage to be 12 V.

(iii) **Collector-base short.** Fig. 6.28 (iii) shows a short between the collector and the base. In this case, the emitter diode is still forward biased so $V_B = 0.7$ V. Now, however, because the collector is shorted to the base, $V_C = V_B = 0.7$ V.

Note. The collector-emitter short is probably the most common type of fault in a transistor. It is because the collector current (I_C) and collector-emitter voltage (V_{CE}) are responsible for the major part of the power dissipation in the transistor. As we shall see (See Art. 6.25), the power dissipation in a transistor is mainly due to I_C and V_{CE} (i.e. $P_D = V_{CE} I_C$). Therefore, the transistor chip between the collector and the emitter is most likely to melt first.

6.13 Characteristics of Common Emitter Connection

The important characteristics of this circuit arrangement are the *input characteristics* and *output characteristics*.

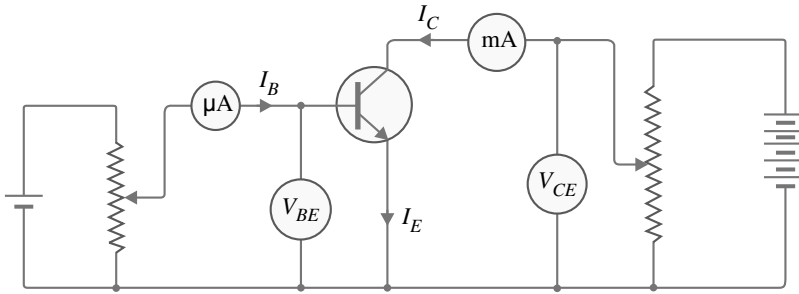


Fig. 6.29

1. Input characteristic. It is the curve between base current I_B and base-emitter voltage V_{BE} at constant collector-emitter voltage V_{CE} .

The input characteristics of a CE connection can be determined by the circuit shown in Fig. 6.29. Keeping V_{CE} constant (say at 10 V), note the base current I_B for various values of V_{BE} . Then plot the readings obtained on the graph, taking I_B along y-axis and V_{BE} along x-axis. This gives the input characteristic at $V_{CE} = 10$ V as shown in Fig. 6.30. Following a similar procedure, a family of input characteristics can be drawn. The following points may be noted from the characteristics :

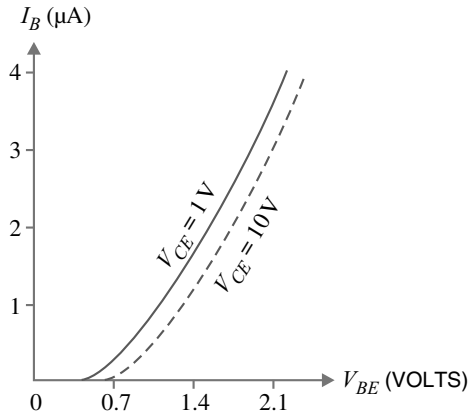


Fig. 6.30

(i) The characteristic resembles that of a forward biased diode curve. This is expected since the base-emitter section of transistor is a diode and it is forward biased.

(ii) As compared to CB arrangement, I_B increases less rapidly with V_{BE} . Therefore, input resistance of a CE circuit is higher than that of CB circuit.

Input resistance. It is the ratio of change in base-emitter voltage (ΔV_{BE}) to the change in base current (ΔI_B) at constant V_{CE} i.e.

$$\text{Input resistance, } r_i = \frac{\Delta V_{BE}}{\Delta I_B} \text{ at constant } V_{CE}$$

The value of input resistance for a CE circuit is of the order of a few hundred ohms.

2. Output characteristic. It is the curve between collector current I_C and collector-emitter voltage V_{CE} at constant base current I_B .

The output characteristics of a CE circuit can be drawn with the help of the circuit shown in Fig. 6.29. Keeping the base current I_B fixed at some value say, $5 \mu\text{A}$, note the collector current I_C for various values of V_{CE} . Then plot the readings on a graph, taking I_C along y-axis and V_{CE} along x-axis. This gives the output characteristic at $I_B = 5 \mu\text{A}$ as

shown in Fig. 6.31 (i). The test can be repeated for $I_B = 10 \mu\text{A}$ to obtain the new output characteristic as shown in Fig. 6.31 (ii). Following similar procedure, a family of output characteristics can be drawn as shown in Fig. 6.31 (iii).

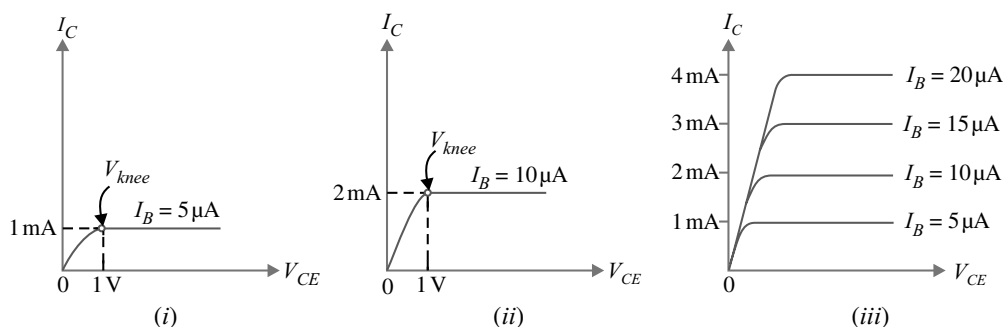


Fig. 6.31

The following points may be noted from the characteristics :

(i) The collector current I_C varies with V_{CE} for V_{CE} between 0 and 1 V only. After this, collector current becomes *almost* constant and independent of V_{CE} . This value of V_{CE} upto which collector current I_C changes with V_{CE} is called the *knee voltage* (V_{knee}). *The transistors are always operated in the region above knee voltage.*

(ii) Above knee voltage, I_C is almost constant. However, a small increase in I_C with increasing V_{CE} is caused by the collector depletion layer getting wider and capturing a few more majority carriers before electron-hole combinations occur in the base area.

(iii) For any value of V_{CE} above knee voltage, the collector current I_C is approximately equal to $\beta \times I_B$.

Output resistance. It is the ratio of change in collector-emitter voltage (ΔV_{CE}) to the change in collector current (ΔI_C) at constant I_B i.e.

$$\text{Output resistance, } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant } I_B$$

It may be noted that whereas the output characteristics of *CB* circuit are horizontal, they have noticeable slope for the *CE* circuit. Therefore, the output resistance of a *CE* circuit is less than that of *CB* circuit. Its value is of the order of 50 k Ω .

Example 6.25. For the transistor circuit shown in Fig. 6.32 (i), determine all node voltages and branch currents. Assume that $V_{BE} = 0.7 \text{ V}$ and $\beta = 100$. Verify that the transistor is operating under normal conditions (i.e. in the active region).

Solution. The circuit shown in Fig. 6.32 (i) can be redrawn as shown in Fig. 6.32 (ii).

$$\text{Emitter voltage, } V_E = 4 - 0.7 = 3.3 \text{ V}$$

$$\text{Emitter current, } I_E = \frac{V_E}{R_E} = \frac{3.3 \text{ V}}{3.3 \text{ k}\Omega} = 1 \text{ mA}$$

$$\text{Collector current, } I_C = \alpha I_E$$

$$\text{Now, } \alpha = \frac{\beta}{\beta + 1} = \frac{100}{100 + 1} = 0.99$$

$$\therefore I_C = 0.99 \times 1 \text{ mA} = 0.99 \text{ mA}$$

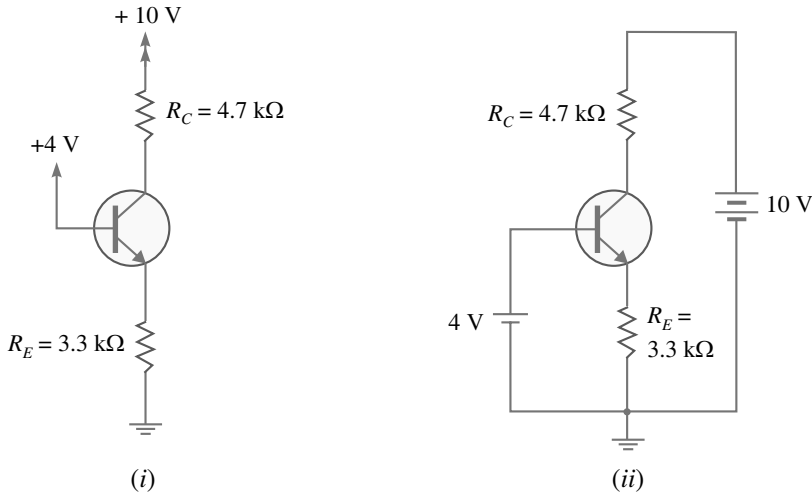


Fig. 6.32

Collector voltage, $V_C = V_{CC} - I_C R_C$

$$= 10 \text{ V} - 0.99 \text{ mA} \times 4.7 \text{ k}\Omega$$

$$= +5.3 \text{ V}$$

Base current, $I_B = I_E - I_C$

$$= 1 - 0.99 = 0.01 \text{ mA}$$

Base voltage, $V_B = 4 \text{ V}$

Since the base is at +4 V, the collector - base junction is reverse biased by 1.3 V ($= 5.3 \text{ V} - 4 \text{ V}$). Further, the base-emitter junction is forward biased because the base is connected to +4 V and the emitter is connected to ground through the resistance R_E . Therefore, the transistor is operating under normal conditions *i.e.* the transistor is operating in the active region.

Fig. 6.33 shows the various node voltages and branch currents. This example will enable the reader to *analyse any transistor circuit with a fair degree of ease.

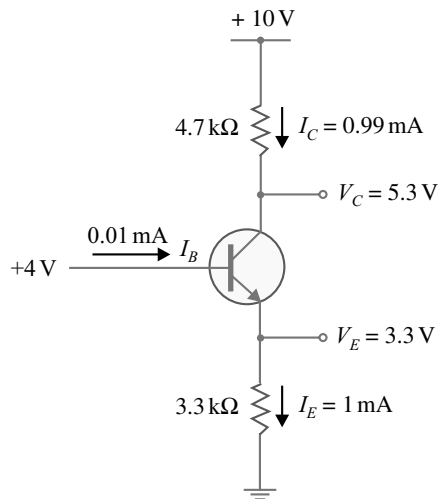


Fig. 6.33

6.14 Common Collector Connection

In this circuit arrangement, input is applied between base and collector while output is taken between the emitter and collector. Here, collector of the transistor is common to both input and output circuits and hence the name common collector connection. Fig. 6.34 (i) shows common collector *nnp* transistor circuit whereas Fig. 6.34 (ii) shows common collector *pnp* circuit.

* Analysing a circuit means finding the various voltages and currents in the circuit.

and current gain. The disadvantage of *CE* circuit is that the leakage current is amplified in the circuit, but bias stabilisation methods can be used.

(iii) **CC circuit.** The input resistance (r_i) and output resistance (r_o) of *CC* circuit are respectively high and low as compared to other circuits. There is no voltage gain ($A_v < 1$) in a *CC* circuit. This circuit is often used for impedance matching.

6.16 Commonly Used Transistor Connection

Out of the three transistor connections, the common emitter circuit is the most efficient. It is used in about 90 to 95 percent of all transistor applications. The main reasons for the widespread use of this circuit arrangement are :

(i) **High current gain.** In a common emitter connection, I_C is the output current and I_B is the input current. In this circuit arrangement, collector current is given by ;

$$I_C = \beta I_B + I_{CEO}$$

As the value of β is very large, therefore, the output current I_C is much more than the input current I_B . Hence, the current gain in *CE* arrangement is very high. It may range from 20 to 500.

(ii) **High voltage and power gain.** Due to high current gain, the common emitter circuit has the highest voltage and power gain of three transistor connections. This is the major reason for using the transistor in this circuit arrangement.

(iii) **Moderate output to input impedance ratio.** In a common emitter circuit, the ratio of output impedance to input impedance is small (about 50). This makes this circuit arrangement an ideal one for coupling between various transistor stages. However, in other connections, the ratio of output impedance to input impedance is very large and hence coupling becomes highly inefficient due to gross mismatching.

6.17 Transistor as an Amplifier in CE Arrangement

Fig. 6.35 shows the common emitter *npn* amplifier circuit. Note that a battery V_{BB} is connected in the input circuit in addition to the signal voltage. This d.c. voltage is known as *bias voltage* and its magnitude is such that it always keeps the emitter-base junction forward *biased regardless of the polarity of the signal source.

Operation. During the positive half-cycle of the **signal, the forward bias across the emitter-base junction is increased. Therefore, more electrons flow from the emitter to the collector *via* the base. This causes an increase in collector current. The increased collector current produces a greater voltage drop across the collector load resistance R_C . However, during the negative half-cycle of the signal, the forward bias across emitter-base junction is decreased. Therefore, collector current decreases. This results in the decreased output voltage (in the opposite direction). Hence, an amplified output is obtained across the load.

* If d.c. bias voltage is not provided, then during negative half-cycle of the signal, the emitter-base junction will be reverse biased. This will upset the transistor action.

** Throughout the book, we shall use sine wave signals because these are convenient for testing amplifiers. But it must be realised that signals (*e.g.* speech, music *etc.*) with which we work are generally complex having little resemblance to a sine wave. However, Fourier series analysis tells us that such complex signals may be expressed as a sum of sine waves of various frequencies.

Transistor Biasing

Learning Objectives

- Develop an understanding of faithful amplification and the conditions to be satisfied by transistor amplifier for this purpose.
- Learn transistor biasing and its need.
- Recognise the need of stabilisation of operating point.
- Understand various methods of transistor biasing.
- Develop a strong understanding of potential-divider bias method.
- Derive an expression of stability factor for potential-divider bias.
- Explore design steps for biasing circuits for low powered transistors.
- Recognise the importance of mid-point biasing of transistor amplifier.
- Learn the comparison of silicon and germanium.

Introduction

The basic function of transistor is amplification. The weak signal is given to the base of the transistor and amplified output is obtained in the collector circuit. One important requirement during amplification is that only the magnitude of the signal should increase and there should be no change in signal shape. This increase in magnitude of the signal without any change in shape is known as *faithful amplification*. In order to achieve this, means are provided to ensure that input circuit (*i.e.* base-emitter junction) of the transistor remains forward biased and output circuit (*i.e.* collector-base junction) always remains reverse biased during all parts of the signal. This is known as transistor biasing. In this chapter, we shall discuss how transistor biasing helps in achieving faithful amplification.

7.1 Faithful Amplification

The process of raising the strength of a weak signal without any change in its general shape is known as faithful amplification.

The theory of transistor reveals that it will function properly if its input circuit (*i.e.* base-emitter junction) remains forward biased and output circuit (*i.e.* collector-base junction) remains reverse biased at all times. This is then the key factor for achieving faithful amplification. To ensure this, the following basic conditions must be satisfied :

- (i) Proper zero signal collector current
- (ii) Minimum proper base-emitter voltage (V_{BE}) at any instant
- (iii) Minimum proper collector-emitter voltage (V_{CE}) at any instant

The conditions (i) and (ii) ensure that base-emitter junction shall remain properly forward biased during all parts of the signal. On the other hand, condition (iii) ensures that base-collector junction shall remain properly reverse biased at all times. In other words, the fulfilment of these conditions will ensure that transistor works over the active region of the output characteristics *i.e.* between saturation to cut-off.

(i) Proper zero signal collector current. Consider an *npn* transistor circuit shown in Fig. 7.1 (i). During the positive half-cycle of the signal, base is positive *w.r.t.* emitter and hence base-emitter junction is forward biased. This will cause a base current and much larger collector current to flow in the circuit. The result is that positive half-cycle of the signal is amplified in the collector as shown. However, during the negative half-cycle of the signal, base-emitter junction is reverse biased and hence no current flows in the circuit. The result is that there is no output due to the negative half-cycle of the signal. Thus we shall get an amplified output of the signal with its negative half-cycles completely cut-off which is unfaithful amplification.

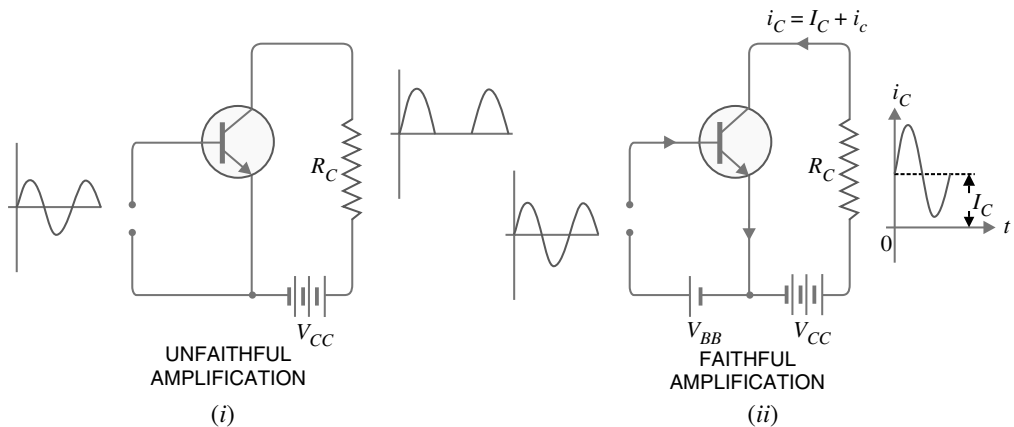


Fig. 7.1

Now, introduce a battery source V_{BB} in the base circuit as shown in Fig. 7.1 (ii). The magnitude of this voltage should be such that it keeps the input circuit forward biased even during the peak of negative half-cycle of the signal. When no signal is applied, a d.c. current I_C will flow in the collector circuit due to V_{BB} as shown. This is known as *zero signal collector current* I_C . During the positive half-cycle of the signal, input circuit is more forward biased and hence collector current increases. However, during the negative half-cycle of the signal, the input circuit is less forward biased and collector current decreases. In this way, negative half-cycle of the signal also appears in the output and hence faithful amplification results. It follows, therefore, that for faithful amplification, proper zero signal collector current must flow. *The value of zero signal collector current should be atleast equal to the maximum collector current due to signal alone i.e.*

Zero signal collector current $\geq$ Max. collector current due to signal alone

Illustration. Suppose a signal applied to the base of a transistor gives a peak collector current of 1mA. Then zero signal collector current must be atleast equal to 1mA so that

even during the peak of negative half-cycle of the signal, there is no cut-off as shown in Fig. 7.2 (i).

If zero signal collector current is less, say 0.5 mA as shown in Fig. 7.2 (ii), then some part (shaded portion) of the negative half-cycle of signal will be cut-off in the output.

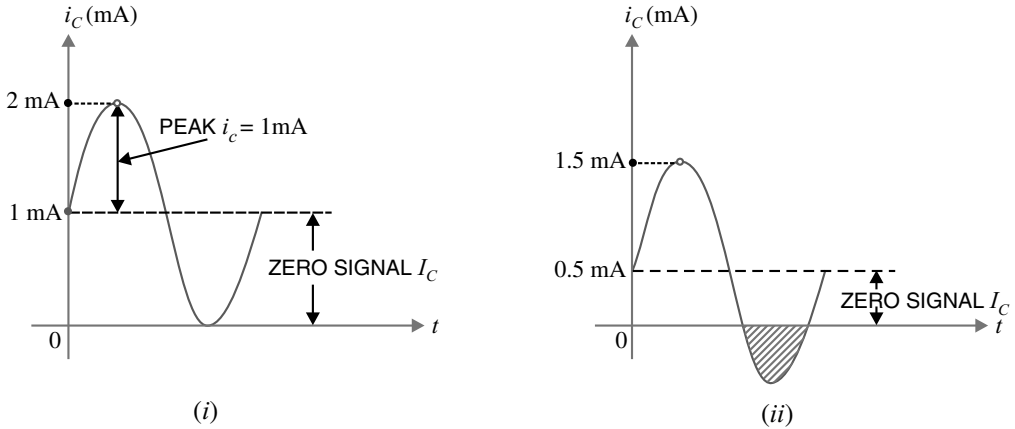


Fig. 7.2

(ii) **Proper minimum base-emitter voltage.** In order to achieve faithful amplification, the base-emitter voltage (V_{BE}) should not fall below 0.5 V for germanium transistors and 0.7 V for Si transistors at any instant.

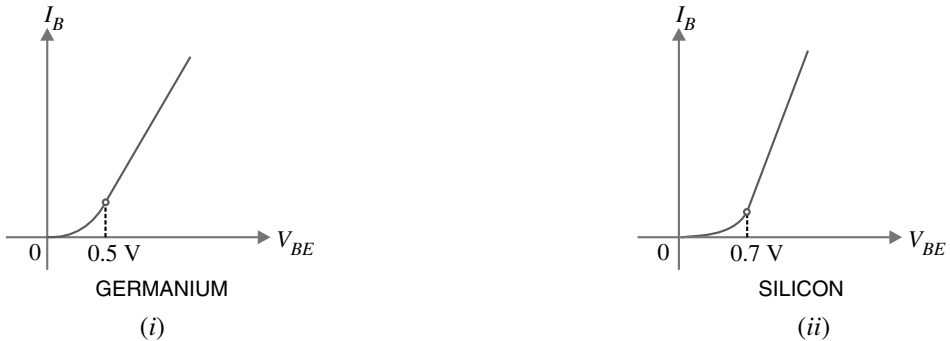


Fig. 7.3

The base current is very small until the *input voltage overcomes the potential barrier at the base-emitter junction. The value of this potential barrier is 0.5 V for Ge transistors and 0.7 V for Si transistors as shown in Fig. 7.3. Once the potential barrier is overcome, the base current and hence collector current increases sharply. Therefore, if base-emitter voltage V_{BE} falls below these values during any part of the signal, that part will be amplified to lesser extent due to small collector current. This will result in unfaithful amplification.

(iii) **Proper minimum V_{CE} at any instant.** For faithful amplification, the collector-emitter voltage V_{CE} should not fall below 0.5 V for Ge transistors and 1 V for silicon transistors. This is called *knee voltage* (See Fig. 7.4).

* In practice, a.c. signals have small voltage level (< 0.1 V) and if applied directly will not give any collector current.

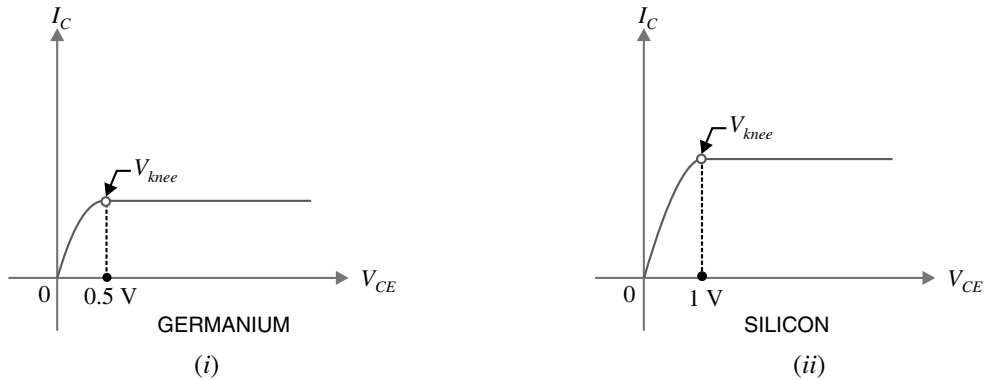


Fig. 7.4

When V_{CE} is too low (less than 0.5 V for Ge transistors and 1 V for Si transistors), the collector-base junction is not properly reverse biased. Therefore, the collector cannot attract the charge carriers emitted by the emitter and hence a greater portion of them goes to the base. This decreases the collector current while base current increases. Hence, value of β falls. Therefore, if V_{CE} is allowed to fall below V_{knee} during any part of the signal, that part will be less amplified due to reduced β . This will result in unfaithful amplification. However, when V_{CE} is greater than V_{knee} , the collector-base junction is properly reverse biased and the value of β remains constant, resulting in faithful amplification.

7.2 Transistor Biasing

It has already been discussed that for faithful amplification, a transistor amplifier must satisfy three basic conditions, namely : (i) proper zero signal collector current, (ii) proper base-emitter voltage at any instant and (iii) proper collector-emitter voltage at any instant. It is the fulfilment of these conditions which is known as transistor biasing.

The proper flow of zero signal collector current and the maintenance of proper collector-emitter voltage during the passage of signal is known as transistor biasing.

The basic purpose of transistor biasing is to keep the base-emitter junction properly forward biased and collector-base junction properly reverse biased during the application of signal. This can be achieved with a bias battery or associating a circuit with a transistor. The latter method is more efficient and is frequently employed. The circuit which provides transistor biasing is known as *biasing circuit*. It may be noted that transistor biasing is very essential for the proper operation of transistor in any circuit.

Example 7.1. An npn silicon transistor has $V_{CC} = 6\text{ V}$ and the collector load $R_C = 2.5\text{ k}\Omega$. Find :

- The maximum collector current that can be allowed during the application of signal for faithful amplification.
- The minimum zero signal collector current required.

Solution. Collector supply voltage, $V_{CC} = 6\text{ V}$

Collector load, $R_C = 2.5\text{ k}\Omega$

- We know that for faithful amplification, V_{CE} should not be less than 1 V for silicon transistor.

7.9 Base Resistor Method

In this method, a high resistance R_B (several hundred $k\Omega$) is connected between the base and +ve end of supply for *nnp* transistor (See Fig. 7.6) and between base and negative end of supply for *pnp* transistor. Here, the required zero signal base current is provided by V_{CC} and it flows through R_B . It is because now base is positive *w.r.t.* emitter *i.e.* base-emitter junction is forward biased. The required value of zero signal base current I_B (and hence $I_C = \beta I_B$) can be made to flow by selecting the proper value of base resistor R_B .

Circuit analysis. It is required to find the value of R_B so that required collector current flows in the zero signal conditions. Let I_C be the required zero signal collector current.

$$\therefore I_B = \frac{I_C}{\beta}$$

Considering the closed circuit $ABENA$ and applying Kirchhoff's voltage law, we get,

$$V_{CC} = I_B R_B + V_{BE}$$

$$\text{or } I_B R_B = V_{CC} - V_{BE}$$

$$\therefore R_B = \frac{V_{CC} - V_{BE}}{I_B} \quad \dots (i)$$

As V_{CC} and I_B are known and V_{BE} can be seen from the transistor manual, therefore, value of R_B can be readily found from exp. (i).

Since V_{BE} is generally quite small as compared to V_{CC} , the former can be neglected with little error. It then follows from exp. (i) that :

$$R_B = \frac{V_{CC}}{I_B}$$

It may be noted that V_{CC} is a fixed known quantity and I_B is chosen at some suitable value. Hence, R_B can always be found directly, and for this reason, this method is sometimes called *fixed-bias method*.

Stability factor. As shown in Art. 7.6,

$$\text{Stability factor, } S = \frac{\beta + 1}{1 - \beta \left(\frac{dI_B}{dI_C} \right)}$$

In fixed-bias method of biasing, I_B is independent of I_C so that $dI_B/dI_C = 0$. Putting the value of $dI_B/dI_C = 0$ in the above expression, we have,

$$\text{Stability factor, } S = \beta + 1$$

Thus the stability factor in a fixed bias is $(\beta + 1)$. This means that I_C changes $(\beta + 1)$ times as much as any change in I_{CO} . For instance, if $\beta = 100$, then $S = 101$ which means that I_C increases 101 times faster than I_{CO} . Due to the large value of S in a fixed bias, it has poor thermal stability.

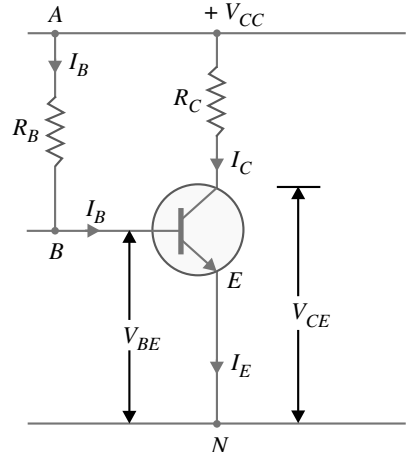


Fig. 7.6

increase in β increases $I_C (= \beta I_B)$. The decrease in V_{BE} increases I_B which in turn increases I_C . As I_C tries to increase, the voltage drop across $R_C (= I_C R_C)$ also tries to increase. This tends to reduce collector voltage V_C (See Fig. 7.25) and, therefore, the voltage across R_B . The reduced voltage across R_B reduces I_B and offsets the attempted increase in I_C and attempted decrease in V_C . The result is that the collector-feedback circuit maintains a stable Q -point. The reverse action occurs when the temperature decreases.

Example 7.22. The collector feedback bias circuit shown in Fig. 7.26 uses a transistor with $\beta = 100$. If the circuit temperature increases from 27°C to 107°C , find the change in collector current. Given that $I_{CBO} = 15 \text{ nA}$ at 27°C .

Solution. The change in collector current (ΔI_C) due to temperature change is

$$\Delta I_C = S \times \Delta I_{CBO}$$

Here, S = Stability factor of the bias circuit

ΔI_{CBO} = Change in I_{CBO} due to temperature change

The stability factor of this bias circuit is

$$S = \frac{\beta + 1}{1 + \beta \left(\frac{R_C}{R_C + R_B} \right)} = \frac{100 + 1}{1 + 100 \left(\frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 270 \text{ k}\Omega} \right)} \approx 56$$

Increase in temp., $\Delta T = 107^\circ\text{C} - 27^\circ\text{C} = 80^\circ\text{C}$

ΔT in 10°C steps, $n = \Delta T / 10^\circ\text{C} = 80^\circ\text{C} / 10^\circ\text{C} = 8$

$$\begin{aligned} \therefore I_{CBO}(\text{new}) &= I_{CBO}(\text{old}) \times 2^n \\ &= 15 \text{ nA} \times 2^8 = 3.84 \text{ }\mu\text{A} \end{aligned}$$

$$\begin{aligned} \text{Now, } \Delta I_{CBO} &= I_{CBO}(\text{new}) - I_{CBO}(\text{old}) \\ &= 3.84 \text{ }\mu\text{A} - 15 \text{ nA} = 3.83 \text{ }\mu\text{A} \end{aligned}$$

$$\therefore \Delta I_C = S \times \Delta I_{CBO} = 56 \times 3.83 \text{ }\mu\text{A} = 214 \text{ }\mu\text{A}$$

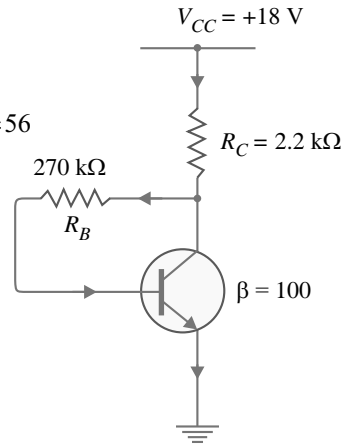


Fig. 7.26

7.13 Voltage Divider Bias Method

This is the most widely used method of providing biasing and stabilisation to a transistor. In this method, two resistances R_1 and R_2 are connected across the supply voltage V_{CC} (See Fig. 7.27) and provide biasing. The emitter resistance R_E provides stabilisation. The name “voltage divider” comes from the voltage divider formed by R_1 and R_2 . The voltage drop across R_2 forward biases the base-emitter junction. This causes the base current and hence collector current flow in the zero signal conditions.

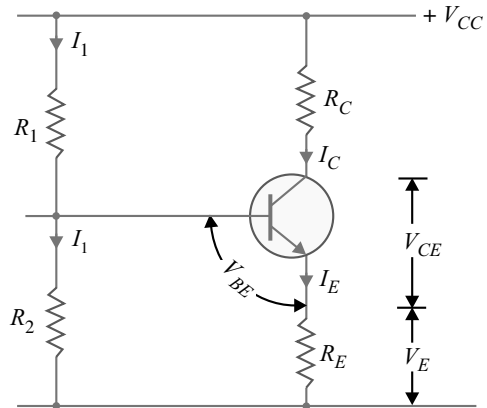


Fig. 7.27

Circuit analysis. Suppose that the current flowing through resistance R_1 is I_1 . As base current I_B is very small, therefore, it can be assumed with reasonable accuracy that current flowing through R_2 is also I_1 .

$$= (100 + 1) \times \frac{1 + 2}{100 + 1 + 2} = 2.94$$

7.15 Design of Transistor Biasing Circuits

(For low powered transistors)

In practice, the following steps are taken to design transistor biasing and stabilisation circuits :

Step 1. It is a common practice to take $R_E = 500 - 1000 \Omega$. Greater the value of R_E , better is the stabilisation. However, if R_E is very large, higher voltage drop across it leaves reduced voltage drop across the collector load. Consequently, the output is decreased. Therefore, a compromise has to be made in the selection of the value of R_E .

Step 2. The zero signal current I_C is chosen according to the signal swing. However, in the initial stages of most transistor amplifiers, zero signal $I_C = 1 \text{ mA}$ is sufficient. The major advantages of selecting this value are :

(i) The output impedance of a transistor is very high at 1 mA . This increases the voltage gain.

(ii) There is little danger of overheating as 1 mA is quite a small collector current.

It may be noted here that working the transistor below zero signal $I_C = 1 \text{ mA}$ is not advisable because of strongly non-linear transistor characteristics.

Step 3. The values of resistances R_1 and R_2 are so selected that current I_1 flowing through R_1 and R_2 is atleast 10 times I_B i.e. $I_1 \geq 10 I_B$. When this condition is satisfied, good stabilisation is achieved.

Step 4. The zero signal I_C should be a little more (say 20%) than the maximum collector current swing due to signal. For example, if collector current change is expected to be 3 mA due to signal, then select zero signal $I_C \approx 3.5 \text{ mA}$. It is important to note this point. Selecting zero signal I_C below this value may cut off a part of negative half-cycle of a signal. On the other hand, selecting a value much above this value (say 15 mA) may unnecessarily overheat the transistor, resulting in wastage of battery power. Moreover, a higher zero signal I_C will reduce the value of R_C (for same V_{CC}), resulting in reduced voltage gain.

Example 7.30. In the circuit shown in Fig. 7.35, the operating point is chosen such that $I_C = 2 \text{ mA}$, $V_{CE} = 3 \text{ V}$. If $R_C = 2.2 \text{ k}\Omega$, $V_{CC} = 9 \text{ V}$ and $\beta = 50$, determine the values of R_1 , R_2 and R_E . Take $V_{BE} = 0.3 \text{ V}$ and $I_1 = 10 I_B$.

Solution. $R_C = 2.2 \text{ k}\Omega$; $V_{CC} = 9 \text{ V}$; $\beta = 50$; $V_{BE} = 0.3 \text{ V}$; $I_1 = 10 I_B$

As I_B is very small as compared to I_1 , therefore, we can assume with reasonable accuracy that I_1 flowing through R_1 also flows through R_2 .

$$\text{Base current, } I_B = \frac{I_C}{\beta} = \frac{2 \text{ mA}}{50} = 0.04 \text{ mA}$$

Current through R_1 & R_2 is

$$I_1 = 10 I_B = 10 \times 0.04 = 0.4 \text{ mA}$$

$$\text{Now, } I_1 = \frac{V_{CC}}{R_1 + R_2}$$

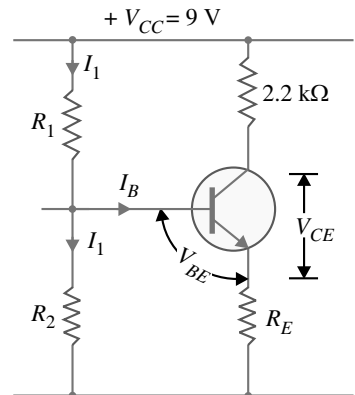


Fig 7.35

The branch of electronics which deals with digital circuits is called **digital electronics**. When most of us hear the term *digital*, we immediately think of “digital calculator” or “digital computer”. This is attributed to the dramatic way the low-cost, powerful calculators and computers have become accessible to an average person. Now digital circuits are being used in many electronic products such as video games, microwave ovens and oscilloscopes. Digital techniques have also replaced a lot of the older “analog circuits” used in consumer products such as radios, TV sets and high-fidelity sound recording and playback equipment. In this chapter, we shall discuss the fundamental aspects of digital electronics.

24.1 Analog and Digital Signals

(i) **Analog signal.** A continuously varying signal (voltage or current) is called an **analog signal**. For example, an alternating voltage varying sinusoidally is an analog signal [See Fig. 24.1]. If such an analog signal is applied to the input of a transistor amplifier, the output voltage will also vary sinusoidally. This is the analog operation *i.e.*, the output voltage can have an infinite number of values. Due to many-valued output, the analog operation is less reliable.

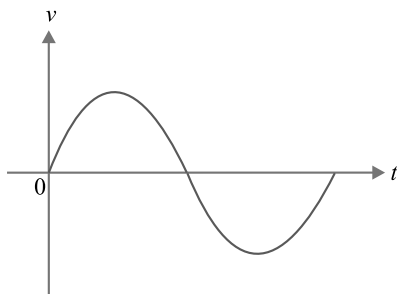


Fig. 24.1

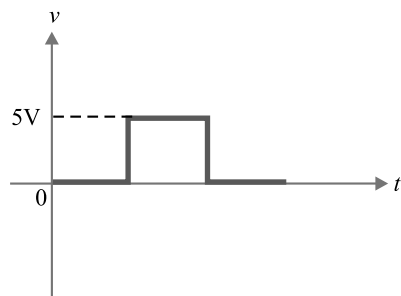


Fig. 24.2

(ii) **Digital signal.** A signal (voltage or current) that can have only two discrete values is called a **digital signal**. For example, a square wave is a digital signal [See Fig. 24.2]. It is because this signal has only two values *viz.*, +5 V and 0 V and no other value. These values are labelled as *High* and *Low*. The High voltage is + 5 V and the Low voltage is 0 V. If proper digital signal is applied to the input of a transistor, the transistor can be driven between *cut-off* and *saturation*. In other words, the transistor will have two-states operation *i.e.*, output is either low or high. Since digital operation has only two states (*i.e.*, *ON* or *OFF*), it is far more reliable than many-valued analog operation. It is because with two-states operation, all the signals are easily recognised as either low or high.

24.2 Digital Circuit

An electronic circuit that handles only a digital signal is called a **digital circuit**.

The output voltage of a digital circuit is either low or high and no other value. In other words, digital operation is a two-states operation. These states are expressed as (*High* or *Low*) or (*ON* or *OFF*) or (1 or 0). Therefore, a digital circuit is one that expresses the values in digits 1's or 0's. Hence the name digital. The numbering concept that uses only the two digits 1 and 0 is the *binary numbering system*. Therefore, the first step would be to discuss the various number systems.

24.3 Number Systems

A number system is a code that uses symbols to count the number of items. The total number of digits used to count number of items in a number system is called **radix or base** of the system. Thus decimal number system uses 10 digits (0 to 9) to count number of items. Therefore, the radix or base of decimal system is 10. There are many number systems but the most common are :

- (i) Decimal number system
- (ii) Binary number system
- (iii) Octal number system
- (iv) Hexadecimal number system

(i) Decimal number system. The most common and familiar number system is the decimal number system. The decimal number system uses 10 digits (0, 1, 2, 3, 4, 5, 6, 7, 8 and 9) for counting items. Therefore, the radix or base of decimal system is 10. The decimal system has evolved naturally as a result of the fact that man has 10 fingers. In fact, the word “digit” is derived from the Latin word for “finger”.

(ii) Binary number system. The binary system uses only two digits (0 and 1) for counting items. Therefore, the radix or base of binary system is 2. Although binary system uses only two digits, yet it can be used to represent any quantity that can be represented in decimal or any other number system.

(iii) Octal number system. The octal number system has a radix of 8 so that it uses eight digits : 0, 1, 2, 3, 4, 5, 6 and 7. Thus each digit of an octal number can have any value from 0 to 7. The ease with which conversions can be made between octal and binary makes the octal system attractive as a “shorthand” means of expressing large binary numbers.

(iv) Hexadecimal number system. The hexadecimal system has a radix of 16. Therefore, it has 16 possible digit symbols. The hexadecimal system uses the first 10 digits of the decimal system (0, 1, 2, 3, 4, 5, 6, 7, 8 and 9) as well as first 6 letters of the alphabet A, B, C, D, E and F. The hexadecimal system is popular in computer applications because large numbers can be stored and the code easily converts to binary.

24.4 Radix or Base of Number System

*The number of digits used in a number system is called **radix or base** of that system.* Thus the radix or base of decimal system is 10 and that of binary system is 2. Similarly, the radix or base of octal system is 8 while that of hexadecimal system is 16.

All number systems are based on the principle of *position weighting* i.e. the value of a digit depends upon its position in the number. For example, consider the decimal number 642. This can be expressed as :

$$642 = 600 + 40 + 2$$

We know that the digit 6 actually represents 6 *hundreds*, the 4 represents 4 *tens* and 2 represents 2 *units*. This decimal number can be written as :

$$642 = 6 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

Thus in a multidigit decimal number, each position has a value that is 10 times (i.e. radix or base times) the value of the next position to its immediate right. In the decimal number under consideration, we find that 6 carries the most weight of the three digits and is referred to as the *most significant digit (MSD)*. The 2 carries the least weight and is called the *least significant digit (LSD)*.

that is 2 times the value of the next position to its immediate right. In other words, every position can be expressed by 2 raised to some power. We know that binary number 1001 is equal to the decimal number 9. This can be readily shown as under :

$$1001 = 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 9$$

For binary numbers, the digit at the extreme right is referred to as *least significant bit (LSB)*. In the binary number 1001, the 1 at the right is the *LSB*. The left-most digit is called the *most significant bit (MSB)*. In the binary number 1001, the 1 at the left is the *MSB*. The *MSB* and *LSB* are illustrated in Fig. 24.5.

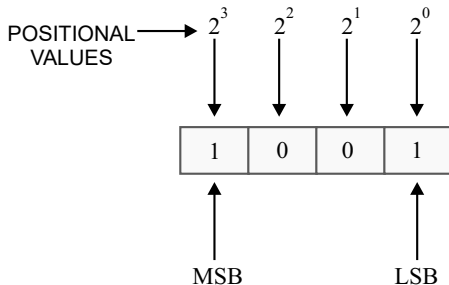


Fig. 24.5

24.7 Decimal to Binary Conversion

Decimal to binary conversion is a more lengthy process. Also there is more than one method that can be used to perform the conversion.

(i) **When decimal number is an integer.** When decimal number is an integer (*i.e.* whole number), we generally use *double dabble method*. It is so called because it requires successive divisions by 2. The method can be summarised as under :

Divide progressively the decimal number by 2 and write down the remainder after each division. Continue this process till you get a quotient of 0 and remainder of 1, the conversion is now complete. The remainders, taken in reverse order form the binary number.

Illustration. Suppose we are to convert decimal number 13 (whole number) into equivalent binary number. The method is illustrated in Fig. 24.6. Note that 13 is first divided by 2, giving a quotient of 6 with a remainder of 1. This remainder becomes the 2^0 position in the binary number. The 6 is then divided by 2, giving a quotient of 3 with a remainder of 0. This remainder becomes the 2^1 position in the binary number.

Continuing this procedure, the equivalent binary number is 1101.

Decimal number

↓

$$\begin{array}{rcl} \boxed{13} & \div & 2 = 6 \\ 6 & \leftarrow \div & 2 = 3 \\ 3 & \leftarrow \div & 2 = 1 \\ 1 & \leftarrow \div & 2 = 0 \end{array}$$

with a remainder of
with a remainder of
with a remainder of
with a remainder of

1
0
1
1

2^0
 2^1
 2^2
 2^3

↑

LSB

MSB

Fig. 24.6

(ii) **When decimal number contains fraction.** When the decimal number contains a fraction (e.g. 1.25), we use two-part method to convert it into equivalent binary number. We first convert the whole part of the decimal number by double dabble method explained above. Next, the fractional part of the decimal number is converted into equivalent binary number by repeatedly multiplying by 2. If the multiplication produces a whole number equal to 1 plus some fractional part, write a 1. The multiplications by 2 are repeated until the fractional part is exactly zero or until a sufficient number of digits have been obtained.

Illustration. Let us convert the decimal number 53.3125 into equivalent binary number. We first convert 53 (whole number part) into equivalent binary number and then convert 0.3125 (fractional part) into equivalent binary number. The two equivalent binary numbers are combined through binary point.

Conversion of 53

2	53	
2	26 - 1	
2	13 - 0	
2	6 - 1	
2	3 - 0	
2	1 - 1	
	0 - 1	

LSB

MSB

$$\therefore (53)_{10} = (110101)_2$$

$$\therefore (53.3125)_{10} = (110101.0101)_2$$

Conversion of 0.3125

0.3125		
$\times 2$		
0.6250	0	MSB
$\times 2$		
1.2500	1	
$\times 2$		
0.5000	0	
$\times 2$		
1.0000	1	LSB

$$\therefore (0.3125)_{10} = (0.0101)_2$$

Example 24.1. Convert $(1245)_{10}$ into binary number system.

Solution. For this conversion, we use double dabble method illustrated below.

2	1245	
2	622 - 1	
2	311 - 0	
2	155 - 1	
2	77 - 1	
2	38 - 1	
2	19 - 0	
2	9 - 1	
2	4 - 1	
2	2 - 0	
2	1 - 0	
	0 - 1	

LSB

MSB

$$\therefore (1245)_{10} = (10011011101)_2$$

Example 24.2. Convert the fractional decimal number $(0.625)_{10}$ into binary number.

Solution. The conversion of fractional decimal number is given below :

0.625		
$\times 2$		
1.25	1	MSB
$\times 2$		
0.50	0	
$\times 2$		
1.00	1	LSB

$$\therefore (0.625)_{10} = (0.101)_2$$

Example 24.3. Convert the decimal number $(100.625)_{10}$ into binary number.

Solution. The conversion is carried out as under :

Conversion of 100

2		100	
2		50 - 0	
2		25 - 0	
2		12 - 1	
2		6 - 0	
2		3 - 0	
2		1 - 1	
		0 - 1	

LSB

MSB

$$\therefore (100)_{10} = (1100100)_2$$

$$\therefore (100.625)_{10} = (1100100.101)_2$$

Conversion of 0.625

0.625		
$\times 2$		
1.25	1	MSB
$\times 2$		
0.50	0	
$\times 2$		
1.00	1	LSB

$$\therefore (0.625)_{10} = (0.101)_2$$

24.8 Binary to Decimal Conversion

Binary numbers can be converted to equivalent decimal numbers quite easily.

(i) **When binary number is an integer.** When binary number is an integer (whole number), its conversion to equivalent decimal number involves the following two steps :

(a) Place the decimal value of each position of the binary number.

(b) Add all the decimal values to get the equivalent decimal number.

Illustration. Suppose we are to convert binary number 110011 (integer) into equivalent decimal number. The binary number along with its decimal values of various positions is shown.

Example 24.13. Convert the binary number $(0.101010110)_2$ to octal number.

Solution. The conversion is done as under :

$$\begin{array}{rccccccc} (0.101010110)_2 = & 0.101 & & 010 & & & 110 \\ & \downarrow & & \downarrow & & & \downarrow \\ & 5 & & 2 & & & 6 \end{array}$$

$$\therefore (0.101010110)_2 = (0.526)_8$$

Example 24.14. Convert $(101011)_2$ into decimal system and octal system.

Solution. The conversion into decimal system is as under :

$$\begin{aligned} (101011)_2 &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 32 + 0 + 8 + 0 + 2 + 1 = 43 \end{aligned}$$

$$\therefore (101011)_2 = (43)_{10}$$

The conversion into octal number is as under :

$$\begin{array}{rccccccc} (101011)_2 = & 101 & & & & 011 & \\ & \downarrow & & & & \downarrow & \\ & 5 & & & & 3 & \end{array}$$

$$\therefore (101011)_2 = (53)_8$$

24.15 Hexadecimal Number System

The hexadecimal system uses a *radix* of 16. Therefore, it has 16 possible digit symbols. The first ten digits in the hexadecimal system are represented by the numbers 0 through 9 (0, 1, 2, 3, 4, 5, 6, 7, 8 and 9) and the letters A through F are used to represent the numbers 10, 11, 12, 13, 14 and 15 respectively. The adjoining table shows the relationships among hexadecimal, decimal and binary. Note that each hexadecimal digit represents a group of four binary digits.

As is true for binary and decimal numbers, each digit in the hexadecimal system has a positional value or weight. For the right most digit of a hex (abbreviation for hexadecimal) number, the positional weight is $16^0 (= 1)$, the next digit to the left has a positional weight of $16^1 (= 16)$ and so on. The positional weight distribution of a hex number system is given below :

$$\begin{array}{ccccccc} & 16^3 & & 16^2 & & 16^1 & & 16^0 \\ \text{etc.} \leftarrow & 4096 & & 256 & & 16 & & 1 \end{array}$$

Hexadecimal	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

1. Decimal-to-hex conversion. To convert a decimal number to hex number, the technique is the same as used for decimal-to-binary conversion or decimal-to-octal conversion. Recall that we did decimal-to-binary conversion using repeated division by 2 and decimal-to-octal conversion using repeated division by 8. Likewise, decimal-to-hex conversion is done using repeated division by 16. Let us illustrate the decimal-to-hex conversion procedure. Suppose we are to convert the decimal number 423 to hex number.

Integer part	Fractional part
16 11011	$0.011 \times 16 = 0.176$ with a carry of 0
16 688 - 3	$0.176 \times 16 = 0.816$ with a carry of 2
16 43 - 0	$0.816 \times 16 = 0.056$ with a carry of 13
16 2 - 11	$0.056 \times 16 = 0.896$ with a carry of 0
0 - 2	$0.896 \times 16 = 0.336$ with a carry of 14

$$\therefore (11011)_{10} = (2B03)_{16} \quad \therefore (0.011)_{10} = (0.02D0E)_{16}$$

$$\therefore (11011.011)_{10} = (2B03.02D0E)_{16}$$

Example 24.22. Convert the decimal number $(39.75)_{10}$ to hexadecimal.

Solution. The conversion is carried out as under :

Integer part	Fractional part
16 39	$0.75 \times 16 = 0$ with a carry of 12 i.e., C
16 2 - 7	
0 - 2	

$$\therefore (39)_{10} = (27)_{16} \quad \therefore (0.75)_{10} = (0.C)_{16}$$

$$\therefore (39.75)_{10} = (27.C)_{16}$$

24.16 Binary-Coded Decimal Code (BCD Code)

Circuits and machines can deal readily with binary numbers, but people are used to working with decimal numbers. Moreover, there are considerably fewer decimal digits required to represent a number than there are binary. It is much easier to remember just a few digits than it is to remember many. Thus whenever there is an interface between digital circuits and people, the interface data usually takes the decimal form. As a result, the digital circuits must utilise some binary code to conveniently represent the decimal numbers. The code used for this purpose is called BCD code. *In a BCD code, each decimal number is represented by a 4-bit binary number.* For example, to convert decimal number $(489)_{10}$ to BCD, the procedure is as under :

4	8	9
0100	1000	1001

Note that the highest BCD value that a 4-bit binary number could represent is 9 which would be $(1001)_2$ in binary. Clearly, only the 4-bit binary numbers from 0000 through 1001 are used.

BCD code	Decimal digit
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9